

Certified Inference and Training for Deep Equilibrium Networks: A Continuation Framework with Polynomial Complexity Guarantees

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Abstract

We develop a certified continuation framework for inference and training in deep equilibrium networks (DEQs), with training posed as interpolation to accuracy 2^{-b} . For inference, input homotopy selects an equilibrium branch from a supplied start root, and a rounded tracker follows it under quantitative conditioning, derivative, boundary, and tube-radius certificates. The framework includes structured factorized certificates, sequential block elimination, inheritance of contraction guarantees in adapted coordinates, and bordered continuation through simple folds. For smooth multidimensional DEQs, including tanh networks, rational local tests can construct and validate oriented continuation charts under explicit geometric promises.

For training, programmable dormant bilinear rank-one channels provide output-preserving residual-aligned repairs. Loaded Tikhonov solves diagnose insufficient parameter-to-output directions, while certified gate realization, column stability, well-posed inference, and finite-update error budgets control each pass. Under polynomially bounded certificate, encoding, precision, and backend costs, both inference and training have bit complexity $O(\text{poly}(L + b))$, where L is the encoded instance length; training uses $O(b + \ell)$ passes and reserve channels from an initial residual bounded by 2^ℓ . A budgeted implementation returns either certified success or inconclusive termination. The quantitative core and local certificate machinery are machine-checked in Lean 4, while numerical experiments illustrate the training mechanism.

1 Introduction

Deep equilibrium networks (DEQs) define a hidden state by an equilibrium equation instead of evaluating a prescribed sequence of layers. This gives an implicit architecture whose effective depth is determined by the equilibrium computation, with training memory advantages that motivated the original DEQ model [8]. The architecture also raises basic theoretical questions: does an equilibrium exist, which equilibrium does the solver select, and how much computation is needed as the requested accuracy increases? Research on monotone architectures, Jacobian regularization, and homotopy methods addresses different parts of these questions [9, 20, 6].

Training adds a second difficulty. Even when every state can be computed reliably, a small change in the parameters may fail to move the predictions in a needed direction. We formulate training as interpolation: for a fixed batch, produce parameters whose prediction vector is within 2^{-b} of the target. The relevant complexity parameter is the number of requested bits b . A guarantee must therefore control both the number of updates and the precision and cost of the calculations inside each update.

Continuation organizes the two computational problems. For inference, we vary the input along a prescribed path and continue an equilibrium from a supplied start root. For training, we prescribe a straight path in batch-output space and seek parameter updates that approximately lift

its displacement. The state Jacobian controls equilibrium continuation; the parameter-to-output Jacobian controls the training lift. The training proof uses finite approximate lifting steps, without assuming an exact continuous parameter path.

As a structured realization of the inference contracts, we study small local recurrent blocks coupled through factors M, C, N . This factorization exposes equilibrium conditioning through an interface matrix, even when the recurrence is non-normal or the usual fixed-point iteration is not contractive. Its dimension need not be small relative to the state dimension for the stated identities and conditioning bounds to hold. For equilibrium computation, input homotopy follows the selected root through a certified region. Our compact continuation theorem and rounded tracker turn bounds on the inverse Jacobian, derivatives, and tube radius into a finite precision schedule and polynomial cost (Theorems 4.6 and 4.7). Section 5 supplies a rational tanh evaluator and an explicit scalar DEQ implementation with a derived polynomial work envelope, rather than an assumed efficient numerical oracle.

For training, a loaded Tikhonov solve either supports a successful interpolation pass or identifies a weak output direction. An unused channel can then be programmed while dormant and activated without changing the current predictions. The important requirement is that its new parameter derivative approximate the current pass residual and remain useful throughout the certified pass region. Detecting a weak direction alone does not ensure that the architecture can realize this repair. Our analysis includes programming error, column variation, inexact evaluation and solves, and rounding. With these contracts, at most $3(b + \ell)$ passes and channels suffice from an initial residual bounded by 2^ℓ (Theorem 7.9).

The resulting claims concern a specified certified promise class. Polynomial computation requires polynomially bounded geometric and precision budgets, efficient gate realization, and numerical backends whose costs include certificate checking. Factorization and smoothness alone do not establish those requirements. This formulation separates the proved quantitative mechanism from the architectural certificates still needed to apply it to a particular DEQ. Lean 4 checks the mathematical core, while the numerical comparisons provide illustrations of the loaded trigger rather than a validation of every certificate. Theorem 4.8 makes the operational outcome explicit: certified success or inconclusive termination within a fixed polynomial budget. Inconclusive does not mean intrinsically hard. Valid old inference certificates retain their original solver and require zero inference repairs. Proposition 4.9 extends coverage by solving contractive recurrent blocks in dependency order; large coupling between blocks costs extra precision bits rather than a polynomial magnitude assumption. This strictly relaxes our uniform Euclidean inverse-bound promise, while drawing on established block and weighted-norm contraction ideas rather than superseding them. Section 4.7 makes certified continuation the default mechanism: an instance certified for an established DEQ solver is solved polynomially by one derivative-free continuation constructed from that solver’s contractive update, including contraction in solver coordinates even when the original state update is not contractive. The outlined exceptions, such as triangular ReLU systems solved by direct substitution, are direct polynomial solutions outside the continuation route. Section 4.8 then extends certified inference beyond the strict-contraction regimes inherited above. At a simple fold two equilibrium branches merge and the update derivative has eigenvalue 1, so no induced-norm strict-contraction certificate can hold at the fold. Crossing folds also changes what inference means: with several coexisting equilibria the requested object is a selected branch, specified by a certified seed, an orientation, and a terminal crossing, and without such a selection rule the input–output relation of a multistable network is not a function. The fold-crossing mechanism itself is classical, and rigorous per-instance continuation exists in validated numerics; the contribution here is certified selected-branch inference for multistable equilibrium networks as a uniform bit-complexity theorem with a machine-checked local layer. A scalar tanh example, the canonical bistable neuron, crosses

two such folds with constant geometric bounds. The extension requires a polynomially bounded branch length and a certified transverse terminal crossing; it is not an unconditional guarantee for arbitrary equilibrium paths. Section 4.10 constructs rational local certificates and assembles overlapping oriented charts without a supplied chart list. Its polynomial guarantee still requires quantitative geometric bounds and a sound terminal-selection monitor. This is an automatic local construction within a stated promise class, not a general decision procedure for equilibrium complexity. The bird’s-eye overview in Section 6 explains the training mechanism before the detailed analysis. Section 8 shows that a successful repair with well-conditioned inference can still require exponentially many updates under the global finite-pass certificate, although a different certified update solves that example efficiently. The final sections discuss composition, experiments, and the scope of the guarantees.

2 Related work and computational complexity

2.1 Equilibrium architectures and inference guarantees

Homotopy-based inference also predates the present construction. Ding et al.’s HomoODE [6] connects DEQs and neural ODEs through homotopy continuation. Beltrán and Leykin [7] give certified rational tracking with condition-dependent bit complexity for polynomial systems. Their polynomial-system theorem cannot be applied verbatim to a network with transcendental activations. Our input-sweep result instead states explicit tube, derivative and evaluation promises; the structured factorization supplies one source of local and interface inverse bounds, without restricting the general continuation theorem to that architecture. This quantitative certificate is the relevant distinction from generic use of a homotopy solver.

The original DEQ formulation computes a weight-tied network’s equilibrium by root finding and differentiates the selected state implicitly, avoiding storage proportional to the effective depth [8]. Multiscale DEQs extend this construction to coupled representations at several resolutions [16]. This memory advantage does not remove the work of the forward root solve or the backward linear solve. In particular, factored inverse updates in Broyden’s method are already part of the DEQ literature; they must be distinguished from the architectural factorized recurrent coupling studied here. Structured connectivity also predates DEQs in recurrent-network theory, where it is used to relate connectivity geometry to low-dimensional dynamics and computation [32]. The present architecture should therefore be assessed through its explicit inference and repair certificates, rather than through the recurrence factorization alone.

Well-posed implicit models also have a substantial history. Implicit deep learning treats fixed-point prediction rules as a common language for feedforward and recurrent architectures and studies sufficient conditions for well-posedness [17]. Monotone operator equilibrium networks construct a parameterization with a unique equilibrium and convergent operator-splitting solvers [9]. NEMON uses non-Euclidean contraction theory to obtain well-posedness, fixed-point algorithms, and quantitative input-output bounds [18]. Thus, guaranteed equilibrium evaluation and stability beyond a naive Euclidean contraction test are established objectives, rather than new claims of this paper.

Scope of the additional inference result. The budgeted formulation in Theorem 4.8 separates sound success, bounded runtime on every input, and coverage of specific certificate classes. A clock and an inconclusive outcome are standard algorithmic devices, not a new way to decide intrinsic hardness. Proposition 4.9 adds explicit binary-precision accounting for sequential contractive blocks: inter-block couplings may have exponential magnitude while their accumulated error amplification

requires only polynomially many extra bits. The previous continuation class is retained as a zero-repair route, including cases that do not satisfy the new block contraction test.

The exact extension proved is over this paper’s uniform Euclidean inverse-bound certificate, as witnessed by Proposition 4.10; it is not a strict containment claim over the classes of El Ghaoui et al. or NEMON [17, 18]. Block structure, changes of norm, and contractive implicit evaluation already belong to that literature. Indeed the separation example also admits a simple weighted contraction certificate. Relative to those antecedents, the result provides an explicit requested-state-accuracy budget, a compositional success or inconclusive interface, and machine-checked preservation and precision lemmas. We do not claim priority for the underlying contraction or elimination mechanism, or a universal automatic certificate generator.

Our inference results specialize this line of analysis to block-local recurrence and a normalized feedback interface. The determinant and inverse reduction use classical matrix identities [15]. The non-normal block transfer estimate likewise has a classical resolvent interpretation. For $A_0 = I - W$ and $0 < d < 1$, write $t = d/(1 - d)$; then

$$d(I - W)(I - dW)^{-1} = tA_0(I + tA_0)^{-1} = I - (I + tA_0)^{-1}.$$

The symmetric-part assumption makes A_0 monotone, and the complementary resolvent is non-expansive by standard monotone-operator estimates [19]. The paper’s role is to combine this estimate with the normalization $N = (I - W_0)^T M$, exposing the reduced denominator and its dimension-independent conditioning margin. These are explicit architectural consequences of established tools, not a new general resolvent theorem. The block-isotropy restriction and the separate equilibrium-selection requirement are essential to that interpretation.

2.2 Reducing the cost of DEQ differentiation

Jacobian regularization improves forward and backward stability at modest additional computational cost [20]. Jacobian-Free Backpropagation (JFB) avoids the implicit Jacobian solve through an alternative update [21], whereas SHINE reuses quasi-Newton inverse estimates from the forward computation to approximate the backward response [22]. More recently, Lipschitz multiscale DEQs study architectural restrictions that guarantee convergence of forward and backward fixed-point computations [30]. A recent preprint on response renormalization targets selected nearly singular, loss-sensitive adjoint response channels [31].

These methods address closely related numerical bottlenecks, but their operators must be distinguished. The state residual Jacobian $J_x = I - DW_{\text{base}}$ governs equilibrium regularity and implicit sensitivities. The training Jacobian $J = DF(\theta)$ maps parameter changes to batch prediction changes. Our normal solve regularizes JJ^* and detects a poorly liftable *training displacement*; a dormant repair changes the available parameter-to-output directions. This differs from approximating or selectively damping the state adjoint while leaving the available training directions unchanged. A bound on J_x^{-1} helps evaluate J and the propagated gate, but does not by itself give a lower bound on the training Jacobian or make that gate surjective.

2.3 Training convergence and the meaning of polynomial complexity

Global training convergence is already known for restricted implicit networks. Kawaguchi proves global linear convergence for implicit layers with nonlinearity on the weight parameterization and relates their dynamics to a trust-region Newton method [23]. Gao et al. analyze over-parameterized ReLU implicit networks [24]. Ling et al. prove linear convergence of gradient descent under quantitative initialization conditions, establish these conditions through an over-parameterization

analysis, and preserve a unique equilibrium throughout training [25]. Truong extends this line of analysis to activation functions with bounded first and second derivatives [26]. Consequently, geometric loss decay or logarithmic dependence on inverse error cannot be claimed as new solely because the model is a DEQ.

Three computational statements should be kept separate. First, a linear algebra operation count measures arithmetic work at a prescribed state and parameter. Second, an iteration bound to reach tolerance ϵ depends on convergence constants, including conditioning and stability margins. Third, polynomial *bit* complexity for $\epsilon = 2^{-b}$ additionally requires polynomially bounded encodings, working precision, and evaluation costs. For a uniform-inverse tracker this includes inverse-margin bounds; block elimination can instead charge the logarithm of inter-block amplification to working precision. For example, contraction with factor κ requires an iteration count proportional to $\log(1/\epsilon)/[-\log \kappa]$; an exponentially small $1 - \kappa$ can invalidate a polynomial bound even though convergence is linear. This observation concerns the resources appearing in a rate, not a contradiction of the cited convergence theorems.

Our result follows the third convention conditionally: the finite-step error budget and repair reserve are explicit, and the complete bit cost is reduced to the certified backend contracts in Assumption 7.8. Since those contracts include an efficient *whole-pass* backend, Theorem 7.9 is a convergence and cost composition theorem; it does not independently construct a polynomial-time trainer for arbitrary DEQs. Establishing such a trainer for a concrete family requires deriving the gate, region, encoding, and backend bounds from that family’s primitive operations.

Classical hardness results for specified neural-network training problems [11] motivate careful formulation of the admissible class. They are not, without a reduction, hardness results for the smooth DEQ architecture or the adaptive reserve considered here. Conversely, well-conditioned promises and architectural growth change the problem being solved; they do not refute those hardness results.

2.4 Function-preserving growth and residual-aligned repair

Function-preserving changes of network architecture precede the present construction: Net2Net transfers a trained function to a wider or deeper network, and network morphism develops a broader framework for such transformations [27, 28]. Factorized parameter updates are also established through adapter methods such as LoRA [10]. In particular, neither output preservation at activation nor a rank-one parameter update is sufficient on its own to establish novelty.

Architectural augmentation also has theoretical optimization guarantees. Liang et al. show, under their classification and loss assumptions, that adding a special neuron together with a regularizer makes every local minimum global [33]. A landscape statement of that kind does not by itself bound the iterations, precision, or work needed to find a minimum. Our finite-pass statement instead concerns an executed trajectory and its quantitative error budgets, conditional on realizable repairs and certified regions.

Lawton, Galstyan, and Ver Steeg use Gauss–Newton approximations to learn and evaluate candidate network morphisms [29]. This is a particularly relevant comparison: second-order guidance of architecture growth is already present in the literature. Our objective is different from ranking candidate growth operations by an approximate decrease in loss. We require a bounded propagated column approximating the fixed pass residual, a derivative-variation bound protecting that column on the pass region, and a numerical budget implying an actual finite-pass contraction. The reserve bound then follows from securing one displacement per pass. It counts activated channels; it does not guarantee realization of arbitrary batch directions by a rank-one gate or bound storage independently of the write dimension.

2.5 Contribution and remaining boundary

For inference, Section 4.7 makes a single continuation mechanism the default route for the quantitative contraction regimes certified in the established solver coordinates, with directly certified exceptions outlined explicitly. This is preservation of their guarantees, not a claim that those methods require the Euclidean inverse bound used by our initial sweep theorem. Section 4.8 separately uses classical pseudo-arclength regularity [2] to pass simple folds. Fold crossing itself is not new: validated continuation certifies branches through folds per instance [1, 2], and homotopy has been used in DEQ inference for acceleration [6]. What appears to be new for equilibrium networks is the problem statement and its complexity: certified selected-branch inference for multistable models beyond strict-contraction regimes. At a fold the update derivative has eigenvalue 1, so no induced-norm strict-contraction certificate can hold there. The result is stated as a uniform bit-complexity theorem over encoded instances with a machine-checked local layer, and composed with rounded tracking, selected-terminal certification, and budgeted termination. Subhomogeneous models [3] are included through their metric contraction only when the coordinate and precision costs are also controlled.

The contribution studied here is the quantitative connection between loaded Tikhonov diagnosis, output-preserving residual-aligned repair, protection under column drift, and finite inexact passes. The mechanism permits a precision-scaled repair reserve without requiring a uniform right inverse of the training Jacobian in every output direction. The factorized inference certificate supplies complementary control of state sensitivities. The machine-checked layer covers the stated abstract implications and their hypotheses; it supports mathematical reliability rather than replacing architectural realizability or experimental validation.

The comparison motivates a certified adaptive-training framework, rather than a claim of the first convergent DEQ trainer, the first use of Gauss–Newton for growth, or unconditional polynomial-time neural-network training. The outstanding step toward a broader algorithmic result is an efficient certificate construction for a nontrivial DEQ family. The historical numerical runs reported later illustrate the intended mechanism but do not certify that step or test all contracts of the finite-pass algorithm.

3 The DEQ architecture and computational problem

For input $u \in \mathbb{R}^{d_u}$, consider

$$x = \varphi(W_0x + MCN^Tx + Uu + c), \quad x \in \mathbb{R}^n. \quad (1)$$

Here W_0 is block diagonal with bounded block sizes, $M, N \in \mathbb{R}^{n \times r}$, and $C \in \mathbb{R}^{r \times r}$. The readout is $\hat{y} = w^Tx + c_o$. Write $W_{\text{base}} = W_0 + MCN^T$ for the fixed recurrent weight used during inference. The coupling dimension r is an explicit computational resource, not a smallness assumption: the factorization identities below hold for any r , and MCN^T can have full rank when $r = n$. A smaller interface dimension can reduce linear-algebra work. This structured model is one way to instantiate the general inference contracts; arbitrary recurrence still requires sound conditioning, domain, and evaluation certificates. The activation acts componentwise and is $C^{1,1}$ on the certified preactivation region. For the monotone activation interpretation we require $0 \leq \varphi' \leq 1$, rather than only $|\varphi'| \leq 1$; tanh is the running example. The abstract training results apply to any $C^{1,1}$ output map with the required evaluation contracts, independently of this derivative range.

Figure 1 shows the local and factorized recurrent contributions and their shared equilibrium feedback.

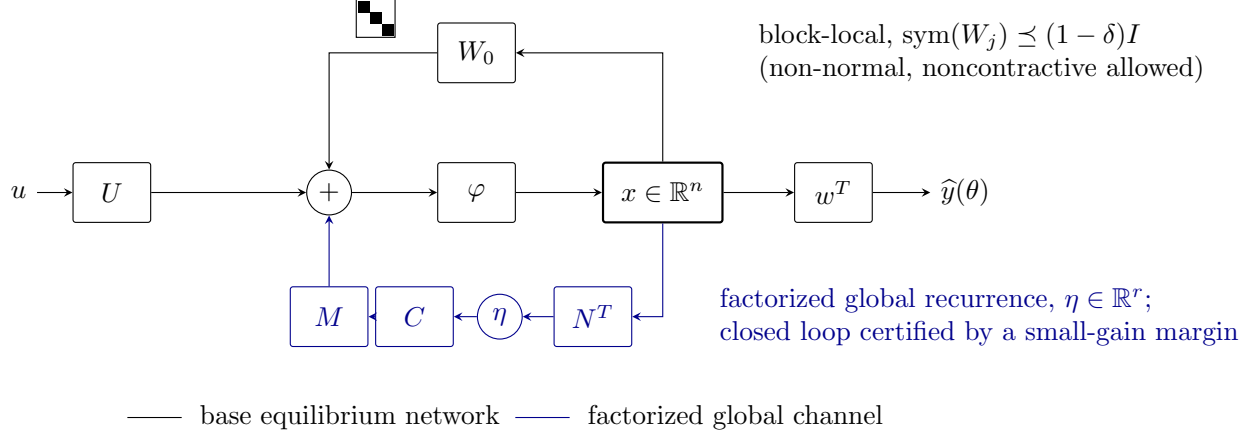


Figure 1: An equilibrium network with factorized recurrence. The equilibrium state solves $x = \varphi(W_0x + MCN^Tx + Uu + c)$. Black: block-local recurrence satisfying the stated symmetric-part bound. Blue: global recurrence through the interface state $\eta = N^Tx$. Under the normalized-family hypotheses, its interface admits a small-gain certificate. The identities hold for any interface dimension r ; a smaller dimension can reduce the cost of the $r \times r$ solve.

For a fixed batch, stack the scalar predictions as $F(\theta) \in \mathbb{R}^m$ and write $J(\theta) = DF(\theta)$. The trainable vector includes the active parameters and any programmed vectors that can subsequently vary.

Given rational instance data and a precision request $b \in \mathbb{N}$, the computational objective is a rational parameter encoding $\hat{\theta}$ with

$$\|F(\hat{\theta}) - y^*\| \leq 2^{-b}. \quad (2)$$

Precision complexity is measured in b , rather than in 2^b . We call this arbitrarily accurate interpolation; no finite exact rational solution is asserted for a general nonlinear network. Parameter dimension is denoted p and the initial-residual exponent by ℓ .

4 Certified inference

The continuation and tracking results concern a general residual map $H(s, x)$ with the stated certificates. We first derive an architectural source of inverse bounds for the factorized model, then state the well-posedness and tracking contracts it can support.

4.1 Exact factorized Jacobian reduction

For the base model (1), let D be the activation derivative matrix. Set

$$A = I - DW_0, \quad B = DMC, \quad L = N^T, \quad Q = I - LA^{-1}B. \quad (3)$$

The state Jacobian is $A - BL$. This definition fixes the ordering of the reduced factors; C need not commute with the reduced transfer.

Theorem 4.1 (Inference reduction). *If A is invertible, then*

$$\det(A - BL) = \det(A) \det(Q).$$

The full and reduced kernels are explicitly isomorphic. If Q is invertible, then

$$(A - BL)^{-1} = A^{-1} + A^{-1}BQ^{-1}LA^{-1}.$$

In particular, bounds $\|A^{-1}\| \leq a$, $\|B\| \leq \beta$, $\|L\| \leq \lambda$, $\|Q^{-1}\| \leq q$ give

$$\|(A - BL)^{-1}\| \leq a + a^2\beta q\lambda.$$

The same inverse-norm bound controls the adjoint sensitivity solve.

Proof. The activation chain rule applied to the state residual $x - \varphi(W_{\text{base}}x + Uu + c)$ gives $J_x = I - DW_{\text{base}}$. Splitting local recurrence and global coupling gives $J_x = A - BL$. Factorization gives

$$A - BL = A(I - A^{-1}BL).$$

The rectangular determinant identity $\det(I - XY) = \det(I - YX)$ with $X = A^{-1}B$, $Y = L$ proves the determinant formula. No commutation of the latent gain with the reduced transfer is used.

If $(A - BL)x = 0$, multiplication by A^{-1} gives $x = A^{-1}BLx$. Therefore $u = Lx$ satisfies $Qu = 0$. Conversely, if $Qu = 0$, then $x = A^{-1}Bu$ satisfies $(A - BL)x = 0$ and $Lx = u$. These maps are inverse linear maps between the kernels; in particular nonzero kernel vectors correspond.

For invertible Q , let $S = A^{-1} + A^{-1}BQ^{-1}LA^{-1}$. Use

$$(I - LA^{-1}B)Q^{-1} = I, \quad Q^{-1}(I - LA^{-1}B) = I$$

to check $(A - BL)S = S(A - BL) = I$ by expanding the products. Submultiplicativity and the triangle inequality then give

$$\|S\| \leq \|A^{-1}\| + \|A^{-1}\|^2 \|B\| \|Q^{-1}\| \|L\| \leq a + a^2\beta q\lambda.$$

The adjoint has the same induced norm, so $\|J_x^{-*}\| = \|J_x^{-1}\|$. □

4.2 Non-normal local blocks

For the normalized family, set $A_0 = I - W_0$, $N = A_0^T M$, $\|M\| \leq 1$, and $C = \alpha I$, with $0 \leq \alpha < 1$. Assume

$$D = \text{blkdiag}(d_j I_{k_j}), \quad 0 \leq d_j \leq 1, \quad \text{sym}(W_j) \preceq (1 - \delta)I, \quad 0 < \delta \leq 1. \quad (4)$$

Block isotropy is a substantive condition. Ordinary elementwise tanh does not satisfy it automatically on multi-coordinate blocks.

Theorem 4.2 (Local transfer and reduced margin). *Under (4), each local Jacobian satisfies $\sigma_{\min}(I - d_j W_j) \geq \delta$ and*

$$\|d_j(I - W_j)(I - d_j W_j)^{-1}\| \leq 1.$$

Consequently $T = A_0(I - DW_0)^{-1}D$ has norm at most one and

$$Q(D) = I - \alpha M^T T M, \quad \sigma_{\min}(Q(D)) \geq 1 - \alpha.$$

No bound $\|W_j\| < 1$ is required.

Proof. For one local block, the strict symmetric-part bound yields

$$\begin{aligned}\langle (I - dW)x, x \rangle &\geq [1 - d(1 - \delta)] \|x\|^2 \\ &\geq \delta \|x\|^2,\end{aligned}$$

where the last inequality uses $0 \leq d \leq 1$ and $0 < \delta \leq 1$. Cauchy–Schwarz gives $\delta \|x\| \leq \|(I - dW)x\|$ for $x \neq 0$, and the inequality is immediate at zero. In finite square dimension this implies invertibility and inverse norm at most δ^{-1} .

The transfer estimate follows from the exact norm identity

$$\begin{aligned}\|x - dWx\|^2 - \|d(x - Wx)\|^2 \\ = (1 - d)^2 \|x\|^2 + 2d(1 - d)(\|x\|^2 - \langle x, Wx \rangle) \geq 0.\end{aligned}$$

Only $\text{sym}(W) \preceq I$ is needed for this estimate. Setting $x = (I - dW)^{-1}y$ proves

$$\|d(I - W)(I - dW)^{-1}y\| \leq \|y\|.$$

The Euclidean norm of a block-diagonal operator is the maximum of its block norms. Since D is scalar on each block, the local operators commute in the required products and $T = A_0(I - DW_0)^{-1}D$ is precisely their block sum. Hence $\|T\| \leq 1$.

With $N = A_0^T M$,

$$Q = I - \alpha M^T A_0 (I - DW_0)^{-1} D M = I - \alpha M^T T M.$$

The compressed transfer $K = M^T T M$ satisfies $\|K\| \leq 1$. For every u ,

$$\|(I - \alpha K)u\| \geq \|u\| - \alpha \|Ku\| \geq (1 - \alpha) \|u\|.$$

This proves the reduced gap and its inverse bound. $\square$

Corollary 4.3 (Dimension-independent conditioning). *If also $\|W_0\| \leq W$, then the full state Jacobian satisfies*

$$\sigma_{\min}(J_x(D)) \geq \frac{\delta^2(1 - \alpha)}{\delta + 1 + W}.$$

At an unsaturated state $D = I$, $J_x = (I - \alpha M M^T) A_0$. If the normalized feedback has a unit eigenvector with eigenvalue one, the reduced gap equals $1 - \alpha$ and there is a unit full-state direction with gain at most $(1 + W)(1 - \alpha)$. Thus, if such a state belongs to the inference region, its worst-case gap is $\Theta(1 - \alpha)$ with constants independent of n and r .

Proof. The local inverse norm is at most δ^{-1} , $\|D\| \leq 1$, $\|M\| \leq 1$, and $\|N\| = \|A_0^T M\| \leq 1 + W$. Theorem 4.1 gives

$$\|J_x^{-1}\| \leq \delta^{-1} + \frac{\alpha(1 + W)}{\delta^2(1 - \alpha)} \leq \frac{\delta + 1 + W}{\delta^2(1 - \alpha)}.$$

Taking reciprocals gives the asserted lower bound.

At $D = I$, the normalization gives $J_x = (I - \alpha M M^T) A_0$ and $Q = I - \alpha M^T M$. If a unit u satisfies $M^T M u = u$, then $\|Q u\| = 1 - \alpha$, which meets the reduced lower bound. Also $y = M u$ is unit and $M M^T y = y$. Put

$$x = \frac{A_0^{-1} y}{\|A_0^{-1} y\|}.$$

Then $\|x\| = 1$ and

$$\|J_x x\| = (1 - \alpha) \|A_0 x\| \leq (1 - \alpha) \|A_0\| \leq (1 + W)(1 - \alpha).$$

If the unsaturated state is present, this upper bound combines with the uniform lower bound to give the worst-case scaling. A unit top feedback eigenvector is a normalization certificate; its existence is not asserted for zero-dimensional feedback or an unnormalized matrix. $\square$

Proposition 4.4 (Anisotropic perturbations). *If a certified state Jacobian J_0 satisfies $\|J_0 v\| \geq g \|v\|$ and the true Jacobian satisfies $\|J - J_0\| \leq \varepsilon < g$, then $\|J v\| \geq (g - \varepsilon) \|v\|$. In finite square dimension it is invertible, with inverse norm at most $(g - \varepsilon)^{-1}$.*

Proof. For every v ,

$$\|J v\| \geq \|J_0 v\| - \|(J - J_0) v\| \geq (g - \varepsilon) \|v\|.$$

The positive lower bound implies injectivity. A linear map on a finite square space is then surjective. Apply the same lower bound to $J^{-1} z$ to obtain the inverse norm estimate. In particular, estimating only the change in the activation derivatives is insufficient unless the induced full Jacobian error is also bounded. $\square$

This permits elementwise activations near a block-isotropic reference when their induced Jacobian perturbation is quantitatively bounded.

4.3 Well-posedness and numerical evaluation

A nonsingular Jacobian is a local regularity certificate. It does not by itself certify a globally unique equilibrium. The training region must also carry a sound well-posedness and evaluation certificate.

Proposition 4.5 (A complete contraction certificate). *If the full layer map $f(x) = \varphi(W_{\text{base}} x + U u + c)$ is κ -contractive, $0 \leq \kappa < 1$, it has exactly one equilibrium x^* and every numerical state $\hat{x}$ satisfies*

$$\|\hat{x} - x^*\| \leq \frac{\|\hat{x} - f(\hat{x})\|}{1 - \kappa}.$$

A sufficient condition is a nonexpansive activation and $\|W_{\text{base}}\| \leq \kappa$.

Proof. The Banach fixed-point theorem applies to the complete Euclidean state space. For a nonexpansive activation,

$$\|f(x) - f(y)\| \leq \|W_{\text{base}}(x - y)\| \leq \kappa \|x - y\|.$$

It gives existence and uniqueness of x^* . The triangle inequality and the contraction estimate give

$$\|\hat{x} - x^*\| \leq \|\hat{x} - f(\hat{x})\| + \kappa \|\hat{x} - x^*\|.$$

Rearranging proves the a posteriori bound. It converts an enclosed layer residual into an equilibrium-state error; the residual of a floating-point layer evaluation must itself include the layer evaluation error. $\square$

The contraction subclass supplies one complete evaluation certificate. Noncontractive instances require another sound state-selection and evaluation backend, such as a quantitatively justified monotone solver; small solve residuals alone are insufficient.

With bounded local block size k , local linear algebra and the reduced solve cost $O(nk^2 + nr + r^3)$ operations after the required local inverses and reduced factors are available. Forming all reduced factors can additionally cost $O(nr^2)$, and applying a precomputed reduced inverse costs $O(r^2)$. Inference iteration counts, factor bit sizes, and requested accuracy remain separate complexity resources. For $r = n$ and bounded k , this accounting gives $O(n^3)$ arithmetic work for the factor formation and solve. The dimensional reduction then offers no size advantage, but the operation count is still polynomial; bit complexity additionally requires the stated numerical and encoding budgets.

4.4 Certified input homotopy beyond contraction

An alternative to contraction is to follow a distinguished equilibrium as the input varies along a supplied sweep $u(s)$, $0 \leq s \leq 1$. Write

$$H(s, x) = x - \varphi(W_{\text{base}}x + Uu(s) + c).$$

All model parameters are fixed during the inference sweep. Its state Jacobian is the same factorized operator used above. A certified positive reduced margin controls the linear systems needed by the tangent predictor and Newton corrector, even when iteration of the layer map is not contractive. The relevant promise concerns the whole certified tube, rather than just the final equilibrium.

Theorem 4.6 (Input-sweep inference certificate). *Let H be C^2 on a neighborhood of $[0, 1] \times \bar{\Omega}$, where $\Omega \subset \mathbb{R}^n$ is bounded and open. Supply a start root $x_0 \in \Omega$, $H(0, x_0) = 0$, and exclude zeros on $[0, 1] \times \partial\Omega$. At every zero in this region assume H_x is invertible and*

$$\|H_x^{-1}\| \leq K, \quad \|H_s\| \leq B.$$

Then the start root has a unique continued branch $x(s)$ for the whole sweep, with

$$x'(s) = -H_x^{-1}H_s, \quad \|x'(s)\| \leq KB.$$

Its graph length is at most $\sqrt{1 + (KB)^2}$. In particular the traversal quantity $\int_0^1 \|H_x^{-1}\| \sqrt{1 + \|x'\|^2} ds$ is at most $K\sqrt{1 + (KB)^2}$.

For numerical tracking, additionally supply a tube of radius $\rho > 0$ around the branch on which the inverse bound holds and the full second derivative is bounded by H_2 . If K, B, H_2, ρ^{-1} , the encoded start data, and evaluation and solve costs at requested precision are polynomially bounded in the input size, a tangent predictor with certified Newton correction tracks this branch to state accuracy 2^{-b} with polynomial total bit cost, provided the evaluator and solver include rounding and certification costs in those bounds.

Proof. The implicit-function theorem gives a local branch and the differentiated identity $H_s + H_x x' = 0$. The inverse bound gives $\|x'\| \leq KB$. If a maximal branch stopped at $s_* < 1$, this velocity bound would make $x(s)$ Cauchy as $s \uparrow s_*$. Its limit lies in $\bar{\Omega}$ and is a zero by continuity. Boundary exclusion puts it in Ω , where the invertible derivative extends the branch, a contradiction. Local uniqueness also makes any two continuations from x_0 agree. Integrating the graph speed proves the length estimates.

For the numerical assertion, differentiating the tangent identity bounds $\|x''\|$ by $KH_2(1 + KB)^2$ using Euclidean product norms. Choose an inverse-polynomial sweep step so that the predictor error is smaller than both a fixed fraction of ρ and a fixed fraction of $(KH_2)^{-1}$; if $H_2 = 0$, only the tube constraint is needed. Taylor's formula bounds this error by a constant times $KH_2(1 + KB)^2 h^2$.

Newton's local error estimate is $\|e_{j+1}\| \leq KH_2 \|e_j\|^2/2$ while its segment remains in the tube. Smaller certified solve and evaluation errors preserve a uniform contraction of the corrector error. An inverse-polynomial step gives polynomially many stages; $O(b)$ certified correction iterations suffice at the terminal stage, with polynomially many precision bits. Summing the supplied bit costs proves the claim. This constructs a tracking schedule under quantitative tube promises; it does not discover them. $\square$

A finite certified tracker. A conservative tracker can use the previous corrected state as its predictor; a tangent predictor is optional. This simpler choice already has polynomial complexity under the tube promises and makes branch selection explicit. Let $V = KB$ and choose $r > 0$ with $r \leq \rho$ and $KH_2 r \leq 1/4$. Use $N \geq 1$ sweep stages with $V/N \leq r/2$, and write $q_i = x(i/N)$. At each stage perform m rounded Newton corrections for $H((i+1)/N, \cdot)$, starting from the preceding corrected state. The branch centers q_i are used in the proof; the algorithm evaluates the equation and its Jacobian, not these unknown exact centers.

Theorem 4.7 (Rounded sweep tracker). *On each radius- r branch ball, suppose the state derivative is H_2 -Lipschitz and has inverse norm at most K . A Newton correction at y uses a step v and a whole-update rounding vector z satisfying*

$$\|H_x v + H\| \leq \eta, \quad \|z\| \leq \zeta, \quad K\eta + \zeta \leq \delta \leq r/8.$$

Here the solve residual includes function and derivative evaluation errors relative to the true equation. Starting at a state within $r/2$ of q_0 , with $m \geq 1$ corrections per stage, every corrected stage endpoint is within $r/2$ of its selected root, and every intermediate correction remains in that root's radius- r ball. For $N \geq 1$ the final state obeys

$$\|\hat{x}_N - q_N\| \leq 2^{-m}r + 2\delta.$$

If $r \leq 2^\ell$, choose $m = b + \ell + 2$ and $\delta \leq \min\{r/8, 2^{-(b+2)}\}$ to obtain state error at most 2^{-b} . There are exactly Nm corrections. If an encoded correction backend simulates these updates, has uniformly polynomial bit cost and output size, and the query size and precision are bounded by S and P , its correction work is at most $Nm A(S + P + 1)^d$ for its fixed family constants A, d . Certification of the tube and stage data has its separately supplied cost.

Proof. For a root q and $e = \|y - q\|$, Taylor's remainder on the convex ball is at most $H_2 e^2$ (a conservative constant). Applying the true inverse to that remainder and the solve residual gives

$$\|y + v + z - q\| \leq KH_2 e^2 + K\eta + \zeta \leq e/4 + \delta \quad (e \leq r).$$

Thus correction stays in the ball; after its first iteration the error is at most $3r/8 < r/2$. Induction also gives $e_j \leq 2^{-j}r + 2\delta$. The branch velocity bound and the mean value theorem give $\|q_{i+1} - q_i\| \leq V/N \leq r/2$. A corrected endpoint therefore starts the next stage within radius r of its root. Induction over stages proves safety and the final estimate. The stated precision schedule bounds the final error by $3 \cdot 2^{-(b+2)} \leq 2^{-b}$.

The encoded and decoded finite recursions agree by induction under the backend simulation contract. Summing the Nm certified correction costs gives the work bound. This identifies the computed state whose error is bounded; a small terminal residual alone would not do so. $\square$

Compactness and local regularity have distinct roles. On the compact zero set in the certified state region, the implicit-function theorem supplies local graph charts for projection to the sweep

interval. These charts make the projection a local homeomorphism; compactness makes it a covering map. Lifting the identity sweep from the start root gives its unique continuous branch. Local agreement with the implicit function gives the derivative formula, including one-sided endpoint neighborhoods. This also explains why a supplied global path is unnecessary.

For the normalized block-isotropic family, Corollary 4.3 supplies $K = (\delta + 1 + W)/(\delta^2(1 - \alpha))$. A nonexpansive activation gives $B \leq \|U\| \sup_s \|u'(s)\|$. Bounded activations also make boundary exclusion inexpensive: for coordinatewise tanh, every root has $\|x\|_\infty < 1$, so the box $(-2, 2)^n$ excludes boundary roots. The reduced-margin and derivative certificates must apply to the actual residual throughout the required region. For general tanh states, block-isotropy is an additional restriction; the anisotropic perturbation certificate can replace it when its slack is positive.

The conclusion selects the branch reached from the supplied start root; it does not exclude other disconnected equilibria. It also gives a controlled warm start for nearby inputs: a certified tube can be reused while its margin and domain tests remain valid. Unlike a generic pseudo-arclength argument, this theorem has a nonsingular H_x throughout, so the input coordinate itself is a valid sweep and no fold bypass is needed. Section 4.8 supplies a separate extension using a bordered margin and a terminal-face certificate in place of these hypotheses, while retaining this original route without repairs.

4.5 Certified success or inconclusive termination

The numerical theorems above are success guarantees on certified classes. They do not decide whether an arbitrary inference problem has intrinsic polynomial complexity. We now place these solvers inside a bounded certification procedure. An inference repair changes the solver, coordinates, or order of elimination while preserving the original equilibrium equation and its selected solution. It does not activate a new recurrent channel and thereby change the network being evaluated.

Fix an encoded backend family, input length S , requested accuracy 2^{-b} , and a polynomial transition budget $P = P(S, b)$. A transition includes its certificate tests and returns either a new state with a repair flag, a certified answer, or *inconclusive*. The wrapper executes at most P transitions; exhaustion also returns inconclusive. The successful-answer predicate for inference is

$$\text{Good}_{\text{inf}}(y) \iff \exists q : q \text{ is the specified equilibrium and } \|\text{decode}(y) - q\| \leq 2^{-b}.$$

For a multivalued equilibrium problem, “specified” includes the continuation branch from the supplied start root. A small residual alone does not establish this predicate. Certificate production may be internal or supplied with the input; its validation and all failed searches are charged to the computation.

Theorem 4.8 (Budgeted certification with preserved coverage). *Suppose the initial state satisfies an invariant, every continuing transition preserves it, and every accepting transition from an invariant state establishes Good_{inf} . Assume that, on all inputs and all transitions reached before the cutoff, each transition has bit cost at most $C(S + b + P + 1)^d$, including certificate search, verification, arithmetic, and rounding. Allow the same bound for initialization and reporting. Then:*

- (i) *The wrapper returns a certified answer or inconclusive, with total bit cost at most $(P + 1)C(S + b + P + 1)^d$. Every returned answer satisfies the specified accuracy and equilibrium-selection condition.*
- (ii) *If the actual strategy has a successful trace of $T \leq P$ transitions with R repair flags, the wrapper returns its same answer after T transitions and reports exactly R repairs.*

- (iii) *In particular, dispatching a valid existing certificate directly to its original tracker, without inference repairs, preserves its answer and has $R = 0$, provided P includes its complete certified work.*

Inconclusive termination makes no assertion about existence, uniqueness, or the intrinsic complexity of the requested equilibrium.

Proof. Define the runner recursively in the remaining budget. At zero it returns inconclusive. Otherwise it executes one transition; a continuing transition recurses with one less unit and adds one to the transition count and its flag to the repair count. Induction bounds transitions by P and repairs by transitions. Induction using invariant preservation and sound acceptance proves answer soundness. A second induction on a successful trace proves that any sufficient budget reproduces the trace, including its answer and repair count. Summing the per-transition bound and one initialization/reporting allowance gives the displayed cost. The zero-repair statement is the trace result with $R = 0$. $\square$

Coverage concerns the executed strategy: the mere existence of a certificate does not ensure that an arbitrary search will find it. We give supplied valid old certificates priority and reserve their full work budget before attempting any optional repair. For a fixed old backend family and a fixed extension family with polynomial coverage bounds, one polynomial P can dominate both bounds and the dispatch overhead. There is no single prescribed polynomial claimed to dominate all possible polynomial-time backend families. The wrapper is standard resource accounting; the additional mathematical content must come from explicit, efficiently checkable coverage classes.

4.6 Block elimination with polynomial precision overhead

We now give an efficiently checkable class whose equilibrium admits polynomial bit complexity even when its global Euclidean inverse bound is exponentially large. After a permutation, let a tanh DEQ have blocks

$$x_i = \tanh \left(W_{ii}x_i + \sum_{j < i} W_{ij}x_j + d_i \right), \quad 1 \leq i \leq m, \quad (5)$$

with rational encoded weights and inputs. Use the induced infinity norm and require $\|W_{ii}\|_\infty \leq 1 - \gamma_i$, where $0 < \gamma_i \leq 1$ and γ_i^{-1} is polynomially bounded in S . Only the *encoding length*, not a polynomial magnitude bound, is required for the off-diagonal couplings. A proposed partition and the norm inequalities are checked with rational arithmetic; alternatively, the directed dependency graph supplies its strongly connected components and a topological ordering in polynomial time. A failed norm test is inconclusive for this route.

Proposition 4.9 (Constructive block coverage). *For this class, sequential certified block solves compute the unique equilibrium to infinity-norm accuracy 2^{-b} with polynomial total bit cost in $S + b$. The block decomposition preserves exactly the solutions of the original equations. It supplies an additional successful route in Theorem 4.8, with at most one decomposition repair followed by ordinary numerical solves. The original certificate route retains priority and uses zero such repairs.*

Proof. Each block map is a contraction for every upstream input, so sequential application of the contraction theorem proves existence and uniqueness. For two upstream inputs, subtract the fixed-point equations, use that tanh is nonexpansive, and move the self-dependence to the left. This gives the sensitivity estimate

$$\|\Delta x_i\|_\infty \leq \gamma_i^{-1} \sum_{j < i} \|W_{ij}\|_\infty \|\Delta x_j\|_\infty.$$

Let $A = \max_i \gamma_i^{-1} \sum_{j < i} \|W_{ij}\|_\infty$. Choose an integer $a \geq 0$ with $1 + A \leq 2^a$, and solve each block to local error $\delta = 2^{-(b+am)}$ for its computed upstream input. The prefix maximum errors E_i satisfy $E_0 = 0$ and

$$E_i \leq \max\{E_{i-1}, \delta + AE_{i-1}\}.$$

Induction gives $E_i \leq \delta(1 + A)^i$: the previous envelope is nondecreasing and the new block error obeys $\delta + A\delta(1 + A)^{i-1} \leq \delta(1 + A)^i$. Consequently,

$$E_m \leq 2^{-(b+am)}(1 + A)^m \leq 2^{-b}. \quad (6)$$

Since rational row sums and γ_i^{-1} have polynomial encoding length, a and $p = b + am$ are polynomially bounded even when A is exponential in magnitude.

Start each block iteration at zero. Its exact initial error is at most one. Set the certified whole-update error to $\nu_i \leq \gamma_i \delta / 2$, including affine evaluation, tanh evaluation, and rounding; coordinatewise clipping to $[-1, 1]$ cannot increase the error to the equilibrium. The inexact contraction recurrence gives after k iterations

$$e_k \leq (1 - \gamma_i)^k + \nu_i / \gamma_i.$$

Thus $k \geq \gamma_i^{-1} \log(2/\delta)$ suffices. This is polynomial. Rational affine operations at these precisions have polynomial cost: large coefficients need their binary length and the additional guard bits, not a number of operations proportional to their magnitude. For a large preactivation z , the bound $1 - |\tanh z| \leq 2e^{-2|z|}$ permits certified saturation to the required precision; otherwise $|z|$ is polynomial in the working precision and standard rational exponential approximation suffices. This gives a polynomial implementation of each certified update. Summing over the blocks proves the bit bound. For Euclidean state accuracy, replace b by $b + \lceil \log_2 n \rceil$, a conservative norm-conversion allowance. $\square$

The general multidimensional tanh block implementation and its primitive bit-cost analysis are separate obligations; they are not identified with the encoded scalar backend.

Proposition 4.10 (Separation from the uniform inverse promise). *The block class contains a family with $O(L)$ -bit descriptions whose Euclidean inverse-Jacobian bound along an input sweep is at least 2^{L+1} , but whose equilibrium is computable in polynomial bit cost in $L + b$.*

Proof. Put $M = 2^L$ and consider

$$x_1 = \tanh(\tfrac{1}{2}x_1 + s), \quad x_2 = \tanh(Mx_1 + s), \quad 0 \leq s \leq 1.$$

At $s = 0$ the unique root is zero, and the residual Jacobian and its inverse are

$$H_x = \begin{pmatrix} 1/2 & 0 \\ -M & 1 \end{pmatrix}, \quad H_x^{-1} = \begin{pmatrix} 2 & 0 \\ 2M & 1 \end{pmatrix}.$$

Applying the inverse to $(1, 0)$ gives norm at least $2M$. Hence this family violates the polynomial- K promise of Theorem 4.6 in the original Euclidean coordinates. The first block has contraction factor $1/2$; compute it to $b + L + O(1)$ bits in $O(b + L)$ iterations, then evaluate the second block. Its propagated error is at most M times the first error plus its own evaluation error. This proves the polynomial bound for rational s of polynomial encoding length, uniformly over the sweep. $\square$

The separation is between sufficient certificates, not an exponential lower bound for every execution of the old tracker. It is also not a separation from weighted-norm contraction methods: the norm $\max\{|x_1|, |x_2|/(4M)\}$ makes the displayed layer map contractive with factor at most $1/2$. Converting back to the requested state norm costs $O(L)$ extra precision bits. Thus the contribution is explicit state-accuracy and bit accounting, together with inclusion of the previous continuation class, rather than a new contraction principle.

4.7 Inheritance of quantitative solver guarantees

The scientific claim of this subsection is that certified continuation is a single default mechanism for certified DEQ inference. First, an instance certified for an established solver—a quantified contraction regime in that solver’s own coordinates—is solved by the one continuation mechanism of Theorem 4.11 with polynomial bit cost, without requiring the original DEQ update to be contractive in the Euclidean state norm. Second, the same mechanism then strictly extends coverage beyond every inherited contraction class: Section 4.8 continues a selected branch through simple folds, where no state-update contraction certificate exists on the route (Example 4.15), and Section 4.10 generates the required local certificates automatically. A small number of deliberate exceptions, such as triangular ReLU systems solved by direct substitution, fall outside the continuation route and are outlined explicitly below. None of this asserts that every convergence theorem is a uniform polynomial bit-complexity theorem.

Theorem 4.11 (Continuation inherited from a contractive solver). *Let T be a contraction with factor $q = 1 - \gamma \in [0, 1)$ on a finite-dimensional normed vector space. Its fixed point must decode to the requested equilibrium. Choose a known y_0 and a computable upper bound $R \geq \max\{1, \|T(y_0) - y_0\|\}$. Define*

$$\Phi(z) = \frac{T(y_0 + Rz) - y_0}{R}, \quad z = s\Phi(z), \quad 0 \leq s \leq 1.$$

Every s has a unique root z_s , with

$$z_0 = 0, \quad \|z_s\| \leq \gamma^{-1}, \quad \|z_s - z_t\| \leq \gamma^{-2}|s - t|.$$

Suppose decoding from y is C -Lipschitz on the encountered region, where $C \geq 1$, and put $p = b + \lceil \log_2(CR) \rceil + 2$. There is a rounded continuation producing output error at most 2^{-b} using

$$O(\gamma^{-3}(p+1))$$

elementary solver evaluations. This is polynomial bit complexity if γ^{-1} , p , encoded trajectory sizes, and the costs of producing or verifying the stated certificates, evaluating Φ , and decoding to the requested precision are polynomial in the input size and b .

Proof. Normalization preserves q and gives $\|\Phi(0)\| \leq 1$. Banach’s theorem applies uniformly to $s\Phi$. From the root equation, $\|z_s\| \leq q\|z_s\| + 1$, and hence also $\|\Phi(z_s)\| \leq \gamma^{-1}$. Subtracting the root equations gives

$$\|z_s - z_t\| \leq q\|z_s - z_t\| + |s - t|\|\Phi(z_t)\|,$$

which proves the motion bound without differentiability assumptions.

Set $r = 1/2$, $N = \lceil 4\gamma^{-2} \rceil$, and $k_0 = \lceil \gamma^{-1} \ln 4 \rceil$. Consecutive roots move by at most $r/2$. A group of k_0 rounded contraction steps, each with additive error at most ν , reduces an initial error e to at most $e/4 + \nu/\gamma$, since $q^{k_0} \leq e^{-\gamma k_0} \leq 1/4$. Choose $\delta \leq \min\{1/16, 2^{-(p+2)}\}$ and $\nu \leq \gamma\delta$. The finite rounded tracker with $m = p + 2$ such groups per stage stays in its tracking balls and ends with normalized error at most 2^{-p} . Decoding amplifies this by at most CR ; an additional output rounding error at most $2^{-(b+1)}$ still leaves total error below 2^{-b} . There are Nmk_0 elementary calls. The bit-cost conclusion counts each call at its required precision; analytic contraction by itself does not supply that cost bound. $\square$

The normalization, coordinate adapters, and source-specific bit backends are separately supplied obligations. This construction preserves polynomiality, not necessarily the original rate or polynomial degree. Intermediate equations are auxiliary solver equations; their endpoint is the original equilibrium. A supplied original certificate may instead use the original zero-repair route directly.

Corollary 4.12 (Explicit coverage of established solver classes). *The following quantitative regimes are covered by the single continuation mechanism of Theorem 4.11, subject to its specified evaluation and encoding requirements.*

1. Perron–Frobenius and block contractions. *For the nonnegative Lipschitz comparison matrix M of the CONE/BLIP conditions of El Ghaoui et al. [17], a certificate $v > 0$, $Mv \leq qv$, $q < 1$, makes the update contractive in the weighted maximum norm (or its block version). Polynomial γ^{-1} and polynomial bit length of the weights give the required regime. For rational M and a supplied rational $q > \rho(M)$, one may compute $v = (qI - M)^{-1}\mathbf{1}$ by rational elimination and check positivity and the inequality. Block norms and activations still require their own certified evaluators. The triangular decomposition results are also inherited through sequential block solves with the precision allocation of Section 4.6.*
2. Monotone-operator DEQs. *For the formulation $0 \in F(x) + \partial f(x)$ of Winston and Kolter [9], let F be m -strongly monotone and L -Lipschitz. Forward–backward splitting with $\alpha = m/L^2$ has factor $q \leq \sqrt{1 - m^2/L^2}$, so $\gamma^{-1} \leq 2(L/m)^2$. For their affine F , Peaceman–Rachford splitting in its auxiliary variable has factor at most $\sqrt{1 - m/L}$ when $\alpha = 1/L$, so $\gamma^{-1} \leq 2L/m$. The proximal and resolvent evaluations and the decoder must have polynomial bit cost; an arbitrary proximal oracle is not a free operation.*
3. Non-Euclidean averaged contractions. *The averaged update in the weighted maximum-norm theorem of Jafarpour et al. [18] has $q = 1 - \alpha(1 - \text{osL}(F))$ for its admissible step sizes. At the stated maximal step, $\gamma^{-1} = \kappa_\infty$. Polynomial effective condition number, encoded norm conversion, and certified evaluation give coverage even when the raw DEQ update is not a Euclidean contraction. The derivative-free theorem also permits nonsmooth activations covered by that source’s hypotheses.*
4. Positive subhomogeneous DEQs. *The positive μ -subhomogeneous maps of Sittoni and Tudisco [3] contract in Thompson distance. With $T(y) = \log F(\exp y)$ the norm is ℓ_∞ and $q = \mu$. For their normalized maps use $q = 2\mu < 1$ in the general case, or $q = \mu < 1$ under their stronger positive-Jacobian hypotheses. Polynomial inverse gap is required. Positivity, the encountered log range, the exponential decoder’s accuracy amplification, and the costs of logarithm and exponential evaluation must also meet Theorem 4.11; the metric change alone does not establish a bit bound.*
5. Architecturally contractive multiscale models. *For a certified forward update with Lipschitz factor $q < 1$, including the contractive construction of Sato and Iiduka [30], the same theorem applies with that factor and with the finite composition’s evaluation cost counted.*

Proof. Weighted norms turn comparison inequalities into ordinary contractions. For forward–backward splitting, nonexpansiveness of the proximal map and strong monotonicity give the squared factor $1 - 2\alpha m + \alpha^2 L^2$. For the reflected affine resolvent, writing $a = F(x) - F(y)$ and $h = x - y$ gives

$$\|h - \alpha a\|^2 = \|h + \alpha a\|^2 - 4\alpha \langle h, a \rangle \leq \left(1 - \frac{4\alpha m}{(1 + \alpha L)^2}\right) \|h + \alpha a\|^2.$$

The other reflection is nonexpansive. Finally, $1 - \sqrt{1 - t} \geq t/2$ for $0 \leq t \leq 1$ yields the stated gaps. The averaged-map rates and Thompson-metric contractions are the cited results, with the latter becoming norm contractions under logarithmic coordinates. Apply Theorem 4.11 in each case. $\square$

Outlined exceptions: direct solutions without a gap. Not every certified class is routed through the homotopy. For triangular scalar ReLU systems the direct formula $x_i = (b_i + \sum_{j < i} A_{ij}x_j)_+ / (1 - A_{ii})$, $A_{ii} < 1$, computes the equilibrium by substitution with polynomial rational cost, including diagonal entries below -1 ; each step adds only polynomially many fraction bits, and no contraction gap exists or is needed. This exception is a polynomial certified solution in its own right and is simply stated as such, not folded into Theorem 4.11.

This list is an explicit coverage statement, not a claim about all past or future DEQ results. For example, geometric convergence with an unspecified instance-dependent factor, such as that established for positive concave DEQs [4], needs a quantitative bound on that factor and its error prefactor before it yields uniform polynomial complexity. An unquantified convergence result is not disproved or excluded by an inconclusive run. Similarly, a noncontractive solver with a separately established polynomial certified implementation is such an exception; it need not fit the contractive homotopy. Absorption here concerns inference, and does not transfer a source’s training or generalization guarantees.

4.8 Continuation through simple folds

The preceding inheritance theorem exploits an existing solver’s contraction. A different extension permits the input to reverse direction while following a selected equilibrium branch. This changes the continuation coordinate, not the equilibrium equation or the architecture. Pseudo-arclength continuation and its regularity at simple folds are classical; see Dickson et al. [2]. The issue here is the quantitative certificate needed to combine that mechanism with finite rounded tracking and budgeted termination.

Here u is a scalar continuation coordinate; a prescribed vector-valued input path may first be composed into H . Write $z = (x, u) \in \mathbb{R}^{n+1}$, $H(u, x) \in \mathbb{R}^n$, and $A = [H_x \ H_u]$. At a regular point of the zero curve let t be an oriented unit vector spanning $\ker A$. The phase-constrained Jacobian is

$$\mathcal{B} = \begin{pmatrix} A \\ t^T \end{pmatrix}.$$

Lemma 4.13 (Bordered margin). *If A has full row rank and smallest positive singular value at least $\sigma > 0$, then $\|\mathcal{B}^{-1}\| \leq \max\{1, \sigma^{-1}\}$. If a frozen-phase Jacobian differs from this one by at most η , its minimum singular value is at least $\min\{1, \sigma\} - \eta$.*

Proof. Decompose $v = v_\perp + at$, with $v_\perp \perp t$. Then $\|\mathcal{B}v\|^2 = \|Av_\perp\|^2 + a^2$ is at least $\min\{1, \sigma^2\} \|v\|^2$. The perturbation statement follows from the triangle inequality. Equivalently, for the normal right inverse $R = A^\dagger$, $\mathcal{B}^{-1}(b, a) = Rb + at$. $\square$

At a simple fold, H_x has a one-dimensional kernel and cokernel. If v, w span them, the conditions $w^T H_u \neq 0$ and $w^T H_{xx}[v, v] \neq 0$ imply a nondegenerate turning point of the input projection. The first condition already makes A full row rank: a left null vector of A would be a multiple of w and would also annihilate H_u . Thus H_x^{-1} may cease to exist while the bordered inverse remains bounded. Qualitative nonzero conditions alone are insufficient for a uniform polynomial bound.

Theorem 4.14 (Quantitative fold extension). *Suppose a selected oriented C^2 zero-curve segment starts at a certified root and reaches a certified terminal bracket for $u = u_*$. Assume the following quantitative promises and backend contracts.*

1. *Its arclength is at most L , A has full row rank with smallest positive singular value at least $\sigma > 0$, and $\|D^2 H\| \leq M$ on a certified neighborhood of radius ρ .*

2. In a terminal neighborhood the selected crossing is isolated by the bracket and $|du/d\ell| \geq \beta > 0$. The bracket specifies which crossing is requested if the branch meets the input face more than once.
3. Local rounded evaluation, linear solving, oriented tangent enclosure, and validation of overlapping phase charts and the terminal bracket have polynomial bit cost at the precisions below. Their encoded states have polynomial size. Validation uses robust margins, with a polynomial bound on failed trials as well as successful trials.

Put $K = \max\{1, \sigma^{-1}\}$ and choose

$$0 < r \leq \min\{1, \rho/4, [64K(1 + M)]^{-1}\}.$$

If $L, K, M, \rho^{-1}, \beta^{-1}, r^{-1}$ and the input/output encoding overheads are polynomially bounded, selected-branch inference admits polynomially many rounded continuation operations, including crossings of simple folds. The number of local charts can be bounded by $O(1 + L/r)$; each has a uniformly bounded inverse and uses polynomial precision and polynomially many corrector steps. The clocked version returns a certified answer or inconclusive. With the original valid certificate given priority, all original successful traces remain available with zero repairs.

Proof. Parametrize the selected segment by arclength. Differentiating $Az' = 0$ and using $\langle z', z'' \rangle = 0$ gives $\|z''\| \leq M/\sigma$. Thus the unit tangent changes by at most KM times the arclength traveled. On a forward arc of length r , the frozen unit tangent t_i has positive inner product, bounded away from zero, with every current tangent. The phase $a = t_i^T(z - z_i)$ is therefore a valid local increasing coordinate. An arclength advance between $r/4$ and r is obtained by choosing a phase advance $r/2$; the same bounds hold with conservative slack for sufficiently accurate enclosures of the center and tangent.

At the center Lemma 4.13 gives inverse bound K . On a ball of radius $2r$, the first row block changes by at most $2Mr$; choose the encoded tangent error at most $1/(16K)$. The perturbed border then has inverse norm at most $2K$. The augmented equations

$$H(z) = 0, \quad t_i^T(z - z_i) - a = 0$$

have derivative in a of norm one. Their local root velocity is at most $2K$, and their derivative variation is bounded by M . Consequently the local contraction radius, phase mesh, and rounding precision can be chosen by the same inequalities as the finite rounded tracker. One can also use rounded Newton correction: Taylor's theorem bounds its exact error by KMe^2 when the inverse is bounded by $2K$, which is at most $e/4$ on these tracking balls; rounding adds δ . Each chart therefore needs only polynomially many corrections at precision b plus logarithmic condition and encoding overheads.

Certified overlaps pass the approximate root and its error enclosure to the next chart on the same oriented local branch. Every completed nonterminal chart advances arclength at least $r/4$, so at most $1 + 4L/r$ charts are required, up to the fixed slack used for enclosures. The local geometric argument establishes available chart size; the assumed backend contract accounts for actually validating and encoding those charts. It is not a free global certificate oracle.

At the terminal face replace the tangent row by the input projection e_u^T . In the decomposition $v = Rb + at$, a terminal right-hand side c requires

$$a = \frac{c - e_u^T Rb}{e_u^T t}, \quad \|v\| \leq K \|b\| + \beta^{-1}(|c| + K \|b\|).$$

This is a polynomial inverse bound. The isolated bracket and its certified neighborhood permit final rounded correction to the requested accuracy and prevent choosing a different crossing. Summing the chart, validation, and terminal costs proves polynomial work. The budgeted wrapper and original-trace preservation follow from Theorem 4.8. $\square$

The length bound and terminal bracket are essential: regularity at every fold does not imply a short branch or that it reaches a specified input. A terminal point which is itself a fold has $\beta = 0$ and needs a different terminal certificate. Rank loss of the full matrix A , as at certain branch points, is also outside this theorem. Coordinate changes may be counted as continuation repairs; they do not add training channels.

Containment of the original inference class. On an original unit-interval input sweep with $\|H_x^{-1}\| \leq K_0$ and $\|H_u\| \leq B_0$, the smallest positive singular value of $[H_x \ H_u]$ is at least K_0^{-1} . Its graph has length at most $\sqrt{1 + (K_0 B_0)^2}$, and its unit tangent satisfies $|du/d\ell| \geq [1 + (K_0 B_0)^2]^{-1/2}$. Thus the old geometric promises imply the new ones after polynomial rescaling. More directly, the dispatcher retains the original backend: membership in the old certified class never forces a fold repair or new chart-validation obligation. The extended accepted class is a union, not a replacement of the original one.

Example 4.15 (A tanh equilibrium curve with two simple folds). Consider $H(u, x) = x - \tanh(2x + u)$. Its zero curve is

$$u(x) = \operatorname{atanh}(x) - 2x, \quad -1 < x < 1.$$

On this curve,

$$H_x = 2x^2 - 1, \quad H_u = -(1 - x^2), \quad H_{xx} = 8x(1 - x^2).$$

At $x = \pm 1/\sqrt{2}$, $H_x = 0$, $H_u = -1/2$, and $H_{xx} = 4x \neq 0$: both are simple folds. The original input-sweep certificate fails there. With phase x , however, the border and its inverse at either fold are

$$\begin{pmatrix} 0 & -1/2 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ -2 & 0 \end{pmatrix}.$$

The Euclidean inverse norm is only 2.

On the whole segment $|x| \leq 4/5$, one has $|H_u| \geq 9/25$, $|H_x| \leq 1$, and $|u'(x)| = |(1 - x^2)^{-1} - 2| \leq 1$. The length is at most $(8/5)\sqrt{2}$. The x -phase inverse sends (h, a) to $(a, (H_x a - h)/(1 - x^2))$ and has maximum-norm operator bound $50/9 < 6$. Thus a single choice of phase crosses both folds with constant geometric bounds. The Hessian of H has norm at most 10, so sufficiently small neighborhoods retain a constant margin. Certified tanh and rational arithmetic provide polynomial evaluation cost in this fixed-dimensional example.

For a concrete selected terminal crossing take $u_* = -51/100$ and $x \in [3/4, 4/5]$. The endpoint values satisfy $u(3/4) < -51/100 < u(4/5)$, and $u'(x) \geq 2/7$ throughout the bracket. For instance $1.94 < \log 7 < 1.95$ and $1.098 < \log 3 < 1.099$ certify these strict inequalities. Such bounds follow from the positive series $\log a = 2 \sum_{j \geq 0} v^{2j+1}/(2j+1)$, $v = (a-1)/(a+1)$, whose remainder after N terms is at most $2v^{2N+1}/((2N+1)(1-v^2))$; $N = 32$ suffices for both displayed intervals. Start at $x = -4/5$, $u = 8/5 - \log 3$, and orient the curve toward increasing x . The initial root is exact in this expression encoding; the logarithm is evaluated with certified rounding. The selected terminal bracket lies beyond both folds and supplies a uniform transversality bound $\beta \geq 2/(7\sqrt{2})$. Hence this is a branch-selection problem to which the fold theorem applies with polynomial cost in b , despite singular H_x along the route. It is a concrete strict extension of the original regular-sweep promise, not a claim that this scalar equation was previously hard.

Corollary 4.16 (Combined inference guarantee). *Fix the original certified backend, a finite set of quantitative solver adapters from Section 4.7, and a fold backend satisfying Theorem 4.14. Require every accepted output to satisfy the same requested equilibrium and branch-selection predicate. With budgets dominating the corresponding polynomial work bounds, their dispatcher succeeds on the union of these promise classes in polynomial bit time. A valid original certificate retains its original zero-repair trace. The fold component strictly enlarges the original regular-sweep promise, as witnessed by Example 4.15.*

Proof. Apply the budgeted dispatcher of Theorem 4.8 to the three routes. The example has a certified selected route through points where H_x is singular, so that route violates the original sweep hypothesis. This is strict containment of certified path problems; it is not a lower bound against other algorithms for the same terminal equation. $\square$

4.9 A finite encoded fold tracker

The geometric extension above now has an executable finite tracking layer. Its input is a finite list of encoded phase charts. Each chart provides a Boolean validation function, an encoded rounded update, an iteration count, a coordinate-repair flag, and cost functions for validation and updating. Roots and analytic certificates occur only in the correctness proof. The validator’s soundness is a contract; arbitrary Boolean acceptance is not a certificate.

The runner charges one tick per validation and correction, and one final tick for successful reporting. It checks each chart before use and executes its correction block only if the block fits the remaining fuel. Failed validation or insufficient fuel returns inconclusive, charging the work already performed. Exhausting the budget at an intermediate state does not produce a certified answer.

Theorem 4.17 (Encoded fold-tracker correctness and bounded work). *Let d decode encoded states into a normed state space. A supplied plan has roots $q_0, \dots, q_N$, radius $r \geq 0$, and rounding allowance $0 \leq \delta \leq r/8$. Assume $\|d(c_0) - q_0\| \leq r/2$ and these chart contracts:*

1. *Adjacent roots satisfy $\|q_i - q_{i+1}\| \leq r/2$. On acceptance of chart i from a valid warm start, its encoded update U_i satisfies, for every encoding c in the target ball,*

$$\|d(U_i(c)) - q_{i+1}\| \leq \frac{1}{4} \|d(c) - q_{i+1}\| + \delta.$$

2. *Every nonterminal chart performs at least one correction. The final chart has m_N corrections satisfying $2^{-m_N} r + 2\delta \leq \varepsilon$. Its root satisfies a predicate Selected including the equilibrium equation, terminal input, and requested branch.*

Every successful answer c satisfies Selected(q_N) and $\|d(c) - q_N\| \leq \varepsilon$. All corrections stay inside their certified target balls. Independently of success, charged ticks and coordinate repairs are at most the fuel budget P .

If every validation succeeds on the actual corrected trace, the runner succeeds with

$$P_* = 1 + \sum_{i=1}^N (1 + m_i).$$

Its repair count is exactly the sum of the chart repair flags, and is zero when all flags are false. If $m_i \leq m$ and every validation and correction has modeled work at most $C \geq 1$, including failed checks, then

$$W_{\text{run}} \leq PC, \quad P_* \leq N(m + 1) + 1.$$

Thus fuel $N(m+1)+1$ suffices on the successful-validation class. Polynomial bounds on N, m, C give polynomial work for this finite layer. For a bit bound, primitive cost functions must cover encoded arithmetic; initialization, certificate construction, and output serialization must have separately counted polynomial budgets.

Proof. The overlap bound turns an $r/2$ warm start for q_i into an r warm start for q_{i+1} . Induction on actual encoded corrections gives $e_k \leq r$ and $e_k \leq 2^{-k}r + 2\delta$. After one or more corrections the error is at most $r/2$, preserving the next warm-start condition. For the final chart use its finer error bound, rather than requiring $r/2 \leq \varepsilon$. Induction over the validated chart list proves selected-terminal accuracy. The statement also permits an empty plan when the initial certified enclosure is already accurate.

The program structurally recurses over the chart list. Each validation consumes one tick, a performed correction block consumes its iteration count, and successful reporting consumes one final tick. Failed checks consume their validation cost. Induction gives the fuel and work bounds even on inconclusive runs. A successful-trace induction proves exact reproduction of corrected states, ticks, and repair counts whenever fuel covers P_* . Summing the iteration bounds proves the uniform budget. $\square$

Connection to bordered Newton correction. For phase functional λ , the chart equation is $\mathcal{F}(z) = (H(z), \lambda(z - z_i) - a)$. Its root identity and the Newton estimate hold even when state and residual spaces have different types. For a bordered derivative J and left inverse R with $\|R\| \leq K$, Taylor remainder bound $H_2 \|z - q\|^2$, solve residual at most η , and rounding error at most ζ , the decoded encoded update satisfies

$$\|z_{\text{new}} - q\| \leq KH_2 \|z - q\|^2 + K\eta + \zeta.$$

Thus $KH_2r \leq 1/4$ and $K\eta + \zeta \leq \delta$ imply the chart contraction contract. The hypothesis connects the actual encoded update to its step and rounding error; it does not identify an arbitrary numerical routine with exact Newton iteration.

Precision without exponentially small charts. Keep the geometric radius r independent of b . If $r \leq 2^\ell$, choose $m_N = b + \ell + 2$ and $\delta \leq \min\{r/8, 2^{-(b+2)}\}$. The checked precision lemma gives $\varepsilon = 2^{-b}$; only precision and correction counts grow with b . Internal charts need only maintain their $r/2$ warm-start accuracy. Shrinking the entire chart radius to 2^{-b} would instead risk exponentially many charts.

An executable fold-crossing instantiation. An executable example instantiates the runner and its semantic theorem for $H(x, u) = x^2 + u$. With phase $x = a$, exact rational Newton is

$$(x, u) \mapsto (a, x^2 - 2ax).$$

In the product maximum norm its new error from $(a, -a^2)$ is $(x - a)^2$, at most one quarter of the previous error on the radius-1/4 ball. This holds for arbitrary rational encoded states, not just the sample trace.

Starting at $(-1/8, -1/64)$, phase targets $-1/16, 0, 1/16, 1/8$ and correction counts $2, 2, 2, 10$ cross the simple fold at $(0, 0)$. The selected endpoint is the positive root at input $u = -1/64$, although the negative start root has that same input. The generic soundness theorem certifies 2^{-8} accuracy; exact evaluation returns $(1/8, -1/64)$. The run uses 21 ticks and one coordinate repair. Fuel 20 yields inconclusive, and an invalid initial enclosure is rejected with its validation charged.

Test costs of three abstract units per check and five per update give 93 units including reporting. These constants test accounting; they are not measured rational-arithmetic bit costs.

The remaining boundary is general geometric certificate construction and its bit-cost analysis, including tangent and terminal-bracket generation. The finite encoded tracker does not establish those separate backend promises.

4.10 Automatic local certificates and selected-path assembly

We now give a construction of local certificates in arbitrary finite dimension, followed by a conditional automatic assembly theorem. The input does not include a list of phase charts. It does include a certified seed, an orientation, quantitative neighborhood bounds, and a terminal-selection contract. These distinctions are essential: producing local certificates is different from discovering an initial equilibrium, bounding the length of its branch, or deciding which endpoint is wanted. Parametrized contraction validation and continuation through folds are established techniques; see [1, 2]. Here we specify their rational tests, search overhead, and connection to the budgeted DEQ tracker.

A rational local test. Use the maximum norm and its induced matrix norm throughout this subsection. Let $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be continuously differentiable on a neighborhood of $B = \{z : \|z - c\|_\infty \leq r\}$, with $c \in \mathbb{Q}^d$ and rational $r > 0$. Choose a rational nonsingular matrix R . Certified enclosures supply

$$Y \geq \|RG(c)\|_\infty, \quad q \geq \sup_{z \in B} \|I - RDG(z)\|_\infty.$$

For example, an entrywise enclosure $DG(B) \subseteq \hat{J} + [-\Delta, \Delta]$ gives the computable rational bound

$$q = \|I - R\hat{J}\|_\infty + \|R\Delta\|_\infty.$$

No sampling of the interior of the box is used.

Lemma 4.18 (Sound rational box certificate). *If $q \leq 1/4$ and $Y \leq r/8$, then G has exactly one zero in B . Its distance from c is at most $Y/(1 - q) \leq r/6$. The map $T(z) = z - RG(z)$ contracts distances by at most q on B .*

Proof. The mean-value estimate gives the contraction bound. Also $\|T(z) - c\|_\infty \leq qr + Y \leq 3r/8$, so T maps the complete closed box into itself. Banach's theorem gives a unique fixed point; nonsingularity of R identifies fixed points with zeros of G . The fixed-point equation gives the stated a posteriori distance bound. $\square$

An accepted test with $q < 1$ already certifies nonsingularity of the preconditioner, so no separate determinant check is charged.

Lemma 4.19 (Local automatic generation). *Suppose $A = DG(c)$ is invertible, $\|A^{-1}\|_\infty \leq K$, $K \geq 1$, and an effective interval evaluator has derivative variation bound Mr on every radius- r box about c with $r \leq \rho$. More precisely, it encloses $DG(c)$ to operator-norm error ϵ_J and the box variation by Mr , with polynomial bit cost in the input length and requested precision. It encloses $G(c)$ to error ϵ_F with the same cost property. Define*

$$r_* = \min\{1, \rho, (64K \max\{1, M\})^{-1}\}.$$

If $\|A^{-1}G(c)\|_\infty \leq r_/128$, an automatic rational search finds a certificate of Lemma 4.18. It needs only polynomially many trials in the input length, $\log K$, and $\log(1/r_*)$, provided the evaluator's encoding overhead is polynomial in these quantities. This assertion does not assume a zero in advance.*

Proof. Choose a dyadic $r \in [r_*/2, r_*]$, an approximation $\hat{A}$ with $\epsilon_J \leq 1/(64K)$, and set $R = \hat{A}^{-1}$ using exact rational elimination. The Neumann estimate gives $\|R\|_\infty \leq 2K$ and $\|RA\|_\infty \leq 2$. The center defect and box variation give $q \leq 2K(\epsilon_J + Mr) \leq 1/16$. With $\epsilon_F \leq r_*/(128K)$, the residual enclosure satisfies

$$Y \leq 2 \|A^{-1}G(c)\|_\infty + 2K\epsilon_F \leq r_*/32 \leq r/16.$$

There is therefore strict slack in both acceptance tests. At stage j , test all radii 2^{-k} , $0 \leq k \leq j$, at precision j , skipping singular approximate matrices. Revisit earlier radii as precision increases. An adequate stage is polynomial in the stated logarithmic quantities, and there are $O(j^2)$ trials up to that stage. Exact rational elimination and the matrix tests have polynomial bit cost. Blindly shrinking a box at a fixed center would not prove completeness: it can eventually exclude the desired zero. $\square$

Explicit enclosures for certified smooth activations. The enclosure layer consumes an activation interface rather than tanh identities. Call a scalar activation *certified* if it supplies a C^1 value φ with derivative $\psi = \varphi'$, a Lipschitz constant M_2 for ψ on the encountered preactivation range, and a rational evaluator computing φ and ψ at rational arguments to accuracy 2^{-p} with polynomial bit cost on that range. For rational data consider

$$F(x, u) = x - \varphi(Wx + vu + a), \quad d = n + 1, \quad s_i = \sum_j |W_{ij}| + |v_i|,$$

with certified activations acting componentwise; heterogeneous per-row activations are permitted, and an affine phase row is the zero activation. The variation of Jacobian row i on a radius- r box has absolute row sum at most $M_{2,i}s_i^2r$. For a phase row with constant derivative, a sharper preconditioned bound is

$$M_R = \max_k \sum_{i=1}^n |R_{ki}| M_{2,i} s_i^2, \quad q \leq q_{\text{center}} + M_R r + \text{enclosure error}.$$

These bounds require polynomially many rational operations. Large coefficients can make the resulting chart radius small; polynomial encoding length alone does not bound the number of charts. For tanh, $|\tanh'| \leq 1$ and $|(1 - \tanh^2)'| \leq 2$ give $M_2 = 2$ and recover the constants $2s_i^2r$. The logistic sigmoid inherits $M_2 = 1/4$ through the exact identity $\sigma(y) = (1 + \tanh(y/2))/2$. The exact Gaussian error linear unit $\text{gelu}(y) = y\Phi(y)$ has second derivative $(2 - y^2)$ times the normal density, and the elementary bounds $|2 - y^2|e^{-y^2/2} \leq 2$ and $\sqrt{2\pi} \geq 2$ give $M_2 = 1$. Their evaluators are supplied by Proposition 4.20 below. Softplus and other smooth activations with bounded second derivative satisfy the interface with their own constants and tail certificates; activations with derivative kinks, such as ReLU, fall outside this lemma and belong to the direct route outlined in Section 4.7. Certified tanh evaluation on arbitrary rational arguments can use saturation when $|y| \geq p + 2$, since $|\tanh y - \text{sign } y| \leq 2e^{-2|y|}$ and $1 - \tanh^2 y \leq 4e^{-2|y|}$. On the remaining interval $|y| = O(p)$, the exponential series with a certified remainder and $O(p)$ terms (with a sufficiently large constant) yields polynomial bit complexity. This extends the implemented bounded evaluator, whose certified bounds cover $|y| \leq 4$. The center enclosures remain hypotheses discharged by each activation's evaluator.

Proposition 4.20 (Sigmoid and GELU evaluation). *(i) For rational u with $|u| \leq 8$ and precision p , the rational values $\hat{\sigma}_p(u) = (1 + \hat{t}_p(u/2))/2$ and $\hat{\sigma}'_p(u) = \hat{d}_p(u/2)/4$ built from the evaluator of Proposition 5.1 satisfy*

$$|\hat{\sigma}_p(u) - \sigma(u)| \leq 2^{-(p+3)}, \quad |\hat{\sigma}'_p(u) - \sigma'(u)| \leq 2^{-(p+3)},$$

with encoded fraction-size and work envelopes equal to those of Proposition 5.1 up to fixed constants. (ii) For rational y and precision p , a rational $\hat{g}_p(y)$ with $|\hat{g}_p(y) - \text{gelu}(y)| \leq 2^{-p}$ is computable with bit cost polynomial in p and the encoding length of y .

Proof. (i) The identity $\sigma(u) = (1 + \tanh(u/2))/2$ halves the $\tanh$ value error and quarters the derivative error; $|u| \leq 8$ places $u/2$ in the certified box, and the transform costs one addition, one constant inversion, and one multiplication beyond the bounded $\tanh$ evaluation. (ii) Write $\text{gelu}(y) = y\Phi(y)$ with $\Phi(y) = \frac{1}{2} + \frac{1}{\sqrt{2\pi}} \int_0^y e^{-t^2/2} dt$. If $y^2 \geq 2(p+2)\ln 2$ and $|y| \geq 1$, the tail bound $\int_{|y|}^\infty e^{-t^2/2} dt \leq e^{-y^2/2}/|y|$ together with $\sqrt{2\pi} \geq 2$ certifies saturation: $\text{gelu}(y)$ differs from y (for $y > 0$) or from 0 (for $y < 0$) by at most $2^{-(p+2)}$, so the saturated value is acceptable and only $|y| = O(\sqrt{p+1})$ remains. There, termwise integration of the exponential series gives the alternating series

$$\int_0^y e^{-t^2/2} dt = \sum_{k \geq 0} \frac{(-1)^k y^{2k+1}}{2^k k! (2k+1)},$$

whose term magnitudes halve once $2(k+1) \geq y^2$; the remainder after $O(p)$ terms (with a fixed constant) is therefore below the first omitted term and at most $2^{-(p+4)}$, by the factorial-decay argument of Proposition 5.1. A rational enclosure of $1/\sqrt{2\pi}$ to $p + O(1)$ bits follows from a certified rational enclosure of π and integer square-root extraction, both classical and polynomial. Multiplying the enclosures and by y , and summing the error contributions weighted by $|y| = O(\sqrt{p+1})$, gives accuracy 2^{-p} after fixed constant adjustments. All expression heights are polynomial in p and the input length, so the integer-fraction envelope of Proposition 5.1 applies verbatim. $\square$

Lemma 4.21 (A finite choice of phase coordinates). *Let $A = DF(z) \in \mathbb{R}^{n \times (n+1)}$ have full row rank and smallest positive singular value at least $\sigma > 0$. At least one of the $d = n+1$ borders $B_j = (A; e_j^T)$ is invertible and satisfies*

$$\|B_j^{-1}\|_\infty \leq 3d \max\{1, \sigma^{-1}\}.$$

Proof. A Euclidean unit vector t spanning $\ker A$ has a component $|t_j| \geq d^{-1/2}$. The solution of $Av = b$, $v_j = s$ is $v = A^\dagger b + t(s - e_j^T A^\dagger b)/t_j$. For $\|(b, s)\|_\infty \leq 1$, its norm is at most $\sigma^{-1}\sqrt{n} + \sqrt{d}(1 + \sigma^{-1}\sqrt{n})$, which is bounded by the displayed quantity. $\square$

Trying all coordinate phases costs d trials per search level, rather than a grid of directions. A simple fold of the input projection does not obstruct this test when DF retains full row rank.

Validating a whole chart family. For a chosen signed coordinate functional $\lambda = \pm e_j^T$, use

$$G_s(z) = (F(z), \lambda(z - c) - s).$$

Its derivative is independent of s . If the same box test gives $q \leq 1/4$ and $Y_0 \leq r/16$, then it validates every $|s| \leq h$ with

$$h = \frac{r}{16 \max\{1, \|Re_d\|_\infty\}}.$$

Indeed $Y_s \leq Y_0 + |s| \|Re_d\|_\infty \leq r/8$. The unique roots $z(s)$ form a continuously differentiable curve by the implicit-function theorem, and

$$\|z'(s)\|_\infty \leq (1 - q)^{-1} \|Re_d\|_\infty, \quad \lambda z'(s) = 1.$$

Thus the certificate describes a connected arc, not just isolated zeros. Derivative enclosures also certify the sign of a previous phase applied to $z'(s)$ at an overlap, fixing the orientation after a

coordinate switch. A quantitative arc bound $(1 - q) \|z(s) - z(s')\|_\infty \leq |s - s'| \|Re_d\|_\infty$ also follows from a contraction argument in place of the differentiated identity.

Theorem 4.22 (Automatic assembly of a selected continuation path). *Let a regular oriented zero-curve segment of the rational DEQ with certified activations, $\tanh$ in particular, start at a supplied certified seed. Suppose the following quantitative promises hold; the segment itself and a chart list are not algorithmic inputs.*

1. *The segment has Euclidean length at most L and lies in a neighborhood with radius ρ , full-row-rank margin σ , and effective derivative enclosure constant M . Coordinates, rational data, and evaluator queries have polynomial encoding length.*
2. *A sound terminal monitor identifies the requested crossing of $u = u_*$ on this oriented branch. It has polynomial cost and a positive detection margin τ : forward arcs of length at most τ cannot skip the selected terminal neighborhood without a certified bracket. Inside that bracket the crossing is unique and $|du/d\ell| \geq \beta > 0$. The monitor can refine enclosures to settle its decisions with polynomial precision. These are explicit terminal-selection promises, not consequences of local regularity.*
3. *$L, \rho^{-1}, \sigma^{-1}, M, \tau^{-1}, \beta^{-1}$ and the complete primitive budgets are polynomially bounded in the instance length. The seed enclosure is refinable with polynomial cost.*

There is a rational adaptive algorithm that generates local box certificates, assembles overlapping oriented charts automatically, and returns the selected equilibrium to accuracy 2^{-b} with polynomial bit cost. No global contraction of the DEQ state update is required. With finite fuel it returns either certified success or inconclusive. The original inference backend can retain priority and its zero-repair trace on every instance already covered by its certificate.

Proof. Set $K_0 = 3d \max\{1, \sigma^{-1}\}$. At each certified root, try the d coordinate borders; Lemma 4.21 provides one with inverse bound K_0 . Shrink a common dyadic radius inside the promised neighborhood so that $K_0 M r$ is smaller than a fixed constant, for example $1/256$. Refining the current root to a rational center costs only $O(\log(1/r) + \log K_0)$ extra bits. Perturbation then preserves a border bound $2K_0$, and further refinement makes its Newton displacement as small as required by Lemma 4.19. The local search therefore finds a certificate with strict slack. Use the same tests with $Y_0 \leq r/16$ to construct the parametric family.

At a transition retain the endpoint enclosure of the previous family. Make it strictly contained in the next box. This enclosure contains an actual zero p on the previously selected arc. Its phase value $s_p = \lambda(p - c)$ is enclosed rationally. Arrange $|s_p| \leq h/8$ by refining that enclosure and the center. The next family's uniqueness identifies $p = z(s_p)$; it cannot select an unrelated zero in the box. Orient the new coordinate using a certified sign of the old phase applied to the new tangent, reversing the sign when necessary. This sign is nonzero at the shared root because both phases parametrize its one-dimensional tangent. Its quantitative margin follows from the bounds on the two bordered inverses: the old unit-tangent component is at least $1/(4\sqrt{d}K_0)$, allowing for variation inside the previous box. Refinement therefore decides the sign with polynomial precision.

Advance to phase $s = h/2$. From $|s_p| \leq h/8$, the forward coordinate displacement is at least $3h/8$, so the traversed arclength is at least $3h/8$. It is at most a fixed multiple of $\sqrt{d}K_0h$ by the derivative bound for the family. Cap h further to make this upper bound at most τ . Uniform neighborhood and conditioning promises permit a common inverse-polynomial lower bound $h_{\min} > 0$ for these accepted advances. The interval certificates have strict slack; dyadic radius and precision searches reach that scale after polynomially many trials, including failures. Thus before terminal

detection there are at most $1 + \lceil 8L/(3h_{\min}) \rceil$ advances. Shared-root uniqueness and the oriented phase changes prove inductively that every advance lies on the branch chosen by the seed and orientation.

Apply the terminal monitor to the validated arcs. Its detection contract gives a bracket for the selected crossing before an advance can skip it. The transverse terminal equation $(F(z), u - u_*) = 0$ has an inverse bounded polynomially in σ^{-1} and β^{-1} , by the tangent/normal decomposition used in Lemma 4.21. Refine the bracket into a validated terminal box, then use rounded contraction corrections to reach 2^{-b} accuracy. Internal radii and progress bounds are independent of b ; terminal correction counts and precision grow polynomially in b . Rational rounding to the certified error budget prevents unbounded denominator growth. Sum matrix, evaluation, search, overlap, monitoring, correction, and output costs to obtain the bit bound. The budgeted dispatcher charges failed attempts as well, and preserving the old backend first gives zero-repair inclusion as before. $\square$

What is automatic, and what remains supplied. The theorem constructs the chart list, local root enclosures, phase changes, and overlap witnesses. It does not infer a global length bound or terminal-selection promise from arbitrary weights. A supplied rational terminal region with certified entry margins and a transverse unique crossing is one way to implement the monitor; automatic discovery of such a region is a separate problem. Without a positive progress scale, even inexpensive local certificates may require exponentially many charts. Full rank loss of DF , an uncertified seed, or an ambiguous terminal event is outside the completeness promise.

5 A concrete rational backend

The numerical contracts can be instantiated constructively for a bounded activation evaluator and a scalar recurrent DEQ. This example separates the cost of representing numerical data from the number of iterations. It does not assume an exact transcendental evaluation oracle.

Proposition 5.1 (Quantitative rational activation evaluation). *For rational z with $|z| \leq 4$ and requested precision $p \in \mathbb{N}$, the Taylor order $n = p + 132$ makes the exponential enclosure at $2|z|$ acceptable. Transform its endpoints through $(e - 1)/(e + 1)$, take their midpoint, and restore the sign of z . The resulting rational value $\hat{t}_p(z)$ and derivative $\hat{d}_p(z) = 1 - \hat{t}_p(z)^2$ satisfy*

$$|\hat{t}_p(z) - \tanh z| \leq 2^{-(p+2)}, \quad |\hat{d}_p(z) - (1 - \tanh^2 z)| \leq 2^{-(p+1)}, \quad |\hat{t}_p(z)| \leq 1, \quad 0 \leq \hat{d}_p(z) \leq 1.$$

For an input fraction whose numerator and denominator have magnitude at most 2^S , an unreduced integer-fraction implementation returns a value fraction with both magnitudes at most $2^{H_p(S)}$, where

$$H_p(S) = C_{\text{enc}}(p + 1)^3(S + 1).$$

Its binary-arithmetic work envelope is at most $C_{\text{eval}}(H_p(S) + 1)^3$. The positive constants C_{enc} and C_{eval} depend only on the fixed activation box and arithmetic model, and are independent of S and p .

Proof. For $|v| \leq 8$, the term $|v|^{32}/32!$ is at most 2^{96} . Every subsequent term is at most half the preceding one. Thus the exponential remainder radius at order $p + 132$ is at most $2^{-(p+3)}$. At the nonnegative argument $2|z|$, the Taylor sum is at least one, so the lower exponential endpoint is positive. The map $(e - 1)/(e + 1)$ has Lipschitz constant at most two on $[0, \infty)$; the midpoint error is consequently at most twice the exponential radius. Squaring values in $[-1, 1]$ amplifies their error by at most two.

The encoding implementation uses signed integer fractions without gcd normalization. Multiplication adds the numerator and denominator size envelopes; addition adds them and charges one further bit. A straight-line expression of height h has fraction size envelope at most h and work envelope $O((h+1)^3)$. The expression for the j th Taylor term has height $O(j(S+1) + (j+1)^2)$: powers contribute linearly in jS , and factorial products contribute quadratically in j . Summing n terms gives height $O(n^2(S+1) + n^3)$; the endpoint transforms change this by a constant factor. Since $n = p + 132 = O(p+1)$, fixed constants give the stated size and work bounds. $\square$

Theorem 5.2 (A finite encoded noncontractive DEQ solver). *For rational u with $|u| \leq 1$, the equilibrium equation*

$$x = \tanh(-2x + u)$$

has a unique root $q \in [-1, 1]$. Given $b \in \mathbb{N}$, set $p = b + 4$, $x_0 = 0$, and compute $b + 2$ updates

$$x_{k+1} = \text{Round}_p \left(\frac{x_k + \hat{t}_p(-2x_k + u)}{2} \right), \quad \text{Round}_p(y) = 2^{-p} \lfloor 2^p y \rfloor.$$

Every stored state belongs to $[-1, 1]$ and has a dyadic representation with numerator magnitude at most 2^p and denominator 2^p . The final state satisfies $|x_{b+2} - q| \leq 2^{-b}$. If the input has fraction size envelope S , there is a constant $C_{\text{run}} > 0$, independent of S , b , and u , such that the encoded run's work envelope satisfies

$$W(S, b) \leq C_{\text{run}}(b+1)(S+b+1)^{12}.$$

In particular, $W(S, b) = O((S+b+1)^{13})$. These are conservative upper bounds for this implementation, not optimality claims.

Proof. For $H_u(x) = x - \tanh(-2x + u)$, one has $1 \leq H'_u(x) \leq 3$. The endpoint signs at -1 and 1 give existence, and strict monotonicity gives uniqueness. Moreover, $|x - q| \leq |H_u(x)|$. The damped map $T_u(x) = (x + \tanh(-2x + u))/2$ has derivative in $[-1/2, 1/2]$, hence contraction factor $1/2$. Its rational approximation preserves the state box. Downward dyadic rounding also preserves this box and has error at most 2^{-p} . The complete update error is at most $2 \cdot 2^{-p}$, giving by induction

$$|x_k - q| \leq 2^{-k} + 4 \cdot 2^{-p}.$$

The stated k and p give the required accuracy. Re-encoding each state as a dyadic fraction prevents the activation evaluator's intermediate denominators from becoming the next state encoding. Each preactivation query has fraction size envelope $O(S+b+1)$ and precision $p = O(b+1)$. The activation expression therefore has height $O((b+1)^3(S+b+1))$, hence at most $O((S+b+1)^4)$. Its cubic work envelope is $O((S+b+1)^{12})$; the quadratic long-division envelope for rounding is absorbed into this bound. Summing over $b+2 = O(b+1)$ actual updates gives the stated cost bound. $\square$

The Jacobian approximation also lies in $[1, 3]$, so the scalar preconditioner $D = 1/2$ passes the rational test $|1 - \hat{H}'_u D| \leq 1/2$ throughout this box. Thus the linear-refinement certificate has a concrete preconditioner here. The original equilibrium map can have derivative magnitude two; the half-damped solver supplies its own contraction certificate.

The work model charges integer arithmetic by conservative schoolbook binary-arithmetic envelopes, including long division for dyadic rounding. The complete update instantiates the same efficient-oracle interface used by the modular results, with derived work and output-size bounds. The formal results verify the envelope bounds and the decoded numerical run; they do not verify native library timing or a low-level bit-machine implementation. This concrete inference family does not discharge multidimensional gate realization or the region certificates required by the adaptive training theorem.

6 The training mechanism from a bird’s-eye view

At the start of a pass, freeze the displacement $\Delta = y^* - F(\theta_0)$ and follow the straight output path $F(\theta_0) + s\Delta$. The pass needs a lift of this one displacement; it need not control every output direction equally.

For training, augment the base recurrent weight by programmable bilinear rank-one channels:

$$W_{\text{eff}} = W_{\text{base}} + \sum_{j=1}^R a_j b_j p_j q_j^T. \quad (7)$$

Both latch coordinates start at zero. Vectors of an unused channel may be programmed before activation while its recurrent contribution remains exactly zero. After a channel is activated, it is part of the fixed recurrent operator. Every subsequent equilibrium evaluation and inference certificate therefore uses the updated residual and state Jacobian, including all activated terms.

Detect the relevant weakness. At a point with training Jacobian J , solve

$$(JJ^* + \tau^2 I)w = \Delta, \quad v = J^*w.$$

If $\|\Delta - Jv\|$ is too large, the normalized solve iterate gives a weak covector with a substantial component of the displacement. Spectral estimation is unnecessary. With approximate solves, explicit residual enclosures and error slack are needed to certify this diagnosis.

Program a residual-aligned repair. For an unused channel, its output derivative is a linear function of its write vector once the read vector is fixed. The trainer finds a bounded program approximating the normalized pass displacement through this gate. It activates $(a_j, b_j) = (\rho, 0)$, preserving the model output. Witness pairing is useful for diagnostics or candidate selection, but cannot replace the realization test. These channels use rank-one adapter directions related to LoRA [10], with a separately programmed activation policy. Figure 2 summarizes the output-preserving activation and the certificates needed before executing a repaired pass.

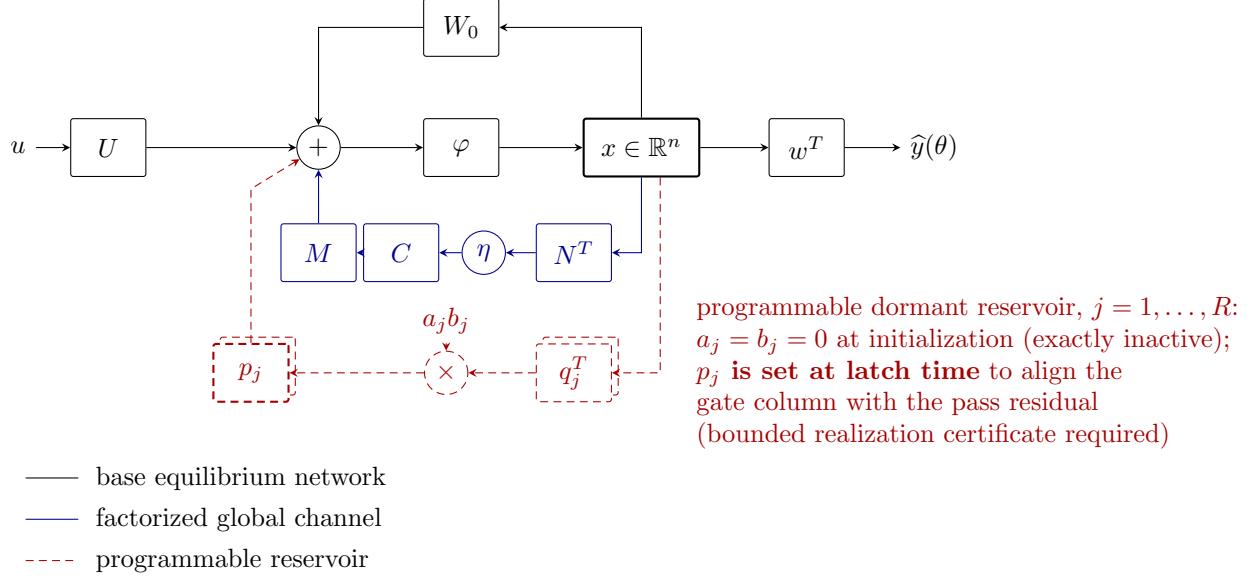


Figure 2: The proposed modification uses the recurrent weight in (7). Red, dashed: the programmable dormant reservoir. While $a_j b_j = 0$ the channel is exactly invisible, so its write vector p_j is not yet data and may legitimately be chosen at activation. A latch $(a_j, b_j): (0, 0) \rightarrow (\rho_*, 0)$ is an exact zero-output operation installing the Jacobian column at activation $\rho_* c_j$; programming p_j aligns c_j with the normalized pass residual. Column drift is controlled on the certified pass region. At most one latch per homotopy pass is ever needed under the realized-pass hypotheses.

Secure one region, not an eternal column. The repair’s effective column is allowed to move. Its initial realization error and its variation over a certified pass ball must fit below the loaded threshold. These bounds secure the fixed displacement on that ball. A later pass uses a new displacement and, if necessary, another unused channel. No permanent global column identity is asserted.

Execute a finite pass. Use N steps with step size $1/N$ and inexact lifted velocities. A speed bound keeps the iterates in the ball. A derivative-Lipschitz bound controls the accumulated truncation error, while the solve and rounding error have their own budget. The resulting pass contracts the output residual by $3/4$. Iteration of this operation gives the precision and reserve count.

Separate success from failed certification. The certified trainer accepts only passes whose region, realization and backend contracts hold. Failure of these checks rejects the pass and its claimed guarantee. A weak loaded covector certifies a current linearized obstruction; reserve exhaustion or a failed region check does not prove global non-realizability of the labels.

7 Training theorems

Let E be the Euclidean parameter space and $Y = \mathbb{R}^m$. Adjoints are with respect to their fixed inner products. All norms below are Euclidean or induced operator norms.

Definition 7.1 (Loaded invariant). *For $\Delta \in Y$ and $\tau > 0$, the Jacobian J has the loaded invariant if*

$$\|u\| = 1, \quad \|J^*u\| \leq \tau \implies 4\langle \Delta, u \rangle^2 < \|\Delta\|^2.$$

Theorem 7.2 (Tikhonov dichotomy). *For the exact regularized normal solve, the velocity satisfies $\tau\|v\| \leq \|\Delta\|$. Either $2\|\Delta - Jv\| \leq \|\Delta\|$, or $w \neq 0$ and $u = w/\|w\|$ satisfies*

$$\|u\| = 1, \quad \|J^*u\| \leq \tau, \quad 2\langle \Delta, u \rangle > \|\Delta\|.$$

In particular the loaded invariant guarantees a lift residual at most $\|\Delta\|/2$.

Proof. Let $A = JJ^*$ and solve $(A + \tau^2 I)w = \Delta$. The operator is invertible:

$$\langle (A + \tau^2 I)z, z \rangle = \|J^*z\|^2 + \tau^2 \|z\|^2,$$

so its kernel is zero, and finite square dimension gives surjectivity. Put $v = J^*w$. The equation gives

$$\Delta - Jv = \tau^2 w, \quad \langle \Delta, w \rangle = \|v\|^2 + \tau^2 \|w\|^2. \quad (8)$$

Cauchy–Schwarz implies $\tau^2 \|w\| \leq \|\Delta\|$ and $\|v\|^2 \leq \|\Delta\| \|w\|$, hence $\tau\|v\| \leq \|\Delta\|$. These claims are immediate if $w = 0$.

For the stronger estimate, expand

$$\|\Delta\|^2 = \|Aw\|^2 + 2\tau^2 \|v\|^2 + \tau^4 \|w\|^2.$$

Since $\|v\|^2 = \langle Aw, w \rangle \leq \|Aw\| \|w\|$ and $(\|Aw\| - \tau^2 \|w\|)^2 \geq 0$, the expansion implies $\|\Delta\|^2 \geq 4\tau^2 \|v\|^2$. Therefore

$$2\tau\|v\| \leq \|\Delta\|. \quad (9)$$

Suppose $2\|\Delta - Jv\| > \|\Delta\|$. By (8), $2\tau^2 \|w\| > \|\Delta\|$, so $w \neq 0$. Normalize $u = w/\|w\|$. Equation (9) gives

$$\|J^*u\| = \|v\|/\|w\| < \tau,$$

and the energy identity gives

$$\langle \Delta, u \rangle = \|v\|^2/\|w\| + \tau^2 \|w\| > \|\Delta\|/2.$$

This is the advertised witness. Squaring its positive loading violates the loaded invariant. No eigendecomposition or smallest-singular-value estimate is used. $\square$

The solve is a regularized least-squares lift, in the tradition of Tikhonov and Levenberg–Marquardt methods [12, 13, 14].

Proposition 7.3 (Certified normal-solve error). *If w is the exact normal-equation solution and $\|(JJ^* + \tau^2 I)\hat{w} - \Delta\| \leq \xi$, then*

$$\|\hat{w} - w\| \leq \xi/\tau^2, \quad \|J^*\hat{w} - J^*w\| \leq \|J\| \xi/\tau^2.$$

The residual bound must concern the true operator and displacement.

Proof. For an exact solution w and approximate $\hat{w}$, let $z = \hat{w} - w$ and $\mathcal{N} = JJ^* + \tau^2 I$. If $\|\mathcal{N}\hat{w} - \Delta\| \leq \xi$, then

$$\tau^2 \|z\|^2 \leq \langle \mathcal{N}z, z \rangle \leq \xi \|z\|, \quad \|z\| \leq \xi/\tau^2.$$

Consequently $\|J^*\hat{w} - J^*w\| \leq \|J\| \xi/\tau^2$. Errors in J , its products, and Δ must first be converted into a bound on the true normal-equation residual. Parameter rounding is then added to the resulting velocity bound.

The exact witness theorem must not be applied without slack to an approximate solve. A direct certified witness check verifies a unit-vector enclosure, the upper bound on $\|J^*u\|$, and the lower bound on its loading. Alternatively, write $\Delta' = \mathcal{N}\hat{w}$ and apply the exact theorem to Δ' , then transfer loading by $|\langle \Delta - \Delta', u \rangle| \leq \xi$. $\square$

Proposition 7.4 (Programming and activation). *Suppose the channel acts through $H((ab)p)$ with H differentiable at zero and derivative G_0 . Programming p while $b = 0$ changes no output. After $a = \rho$, the b -derivative is $\rho G_0 p$. Write $G = \rho G_0$ for the full post-latch gate. If a numerical gate $\hat{G}$ and a program p satisfy*

$$\|\rho d - \hat{G}p\| \leq \eta_g, \quad \|G - \hat{G}\| \leq \gamma, \quad \|p\| \leq S,$$

then $\|\rho d - Gp\| \leq \eta_g + \gamma S$.

Proof. While $b = 0$, $H((ab)p) = H(0)$ for every a, p . Both programming and activation preserve the output exactly as algebraic operations. After activation, differentiation of $b \mapsto H((\rho b)p)$ at zero gives $DH(0)[\rho p] = \rho G_0 p = Gp$. The gate approximation bound follows from

$$\|\rho d - Gp\| \leq \|\rho d - \hat{G}p\| + \|(G - \hat{G})p\| \leq \eta_g + \gamma S.$$

For the recurrent adapter, fix its read vector q and let x_i be the selected equilibrium for sample i . With invertible state Jacobian $A_i = I - D_i W_{\text{eff}}$, its unscaled write-side gate has rows

$$(G_q p)_i = (q^T x_i) w^T A_i^{-1} D_i p. \quad (10)$$

This formula follows by differentiating the equilibrium equation with respect to a recurrent perturbation pq^T . It is linear in p but depends on the states, readout and local inverse. The latent inference certificate controls the inverse in this formula; it does not make G_q surjective.

The adjoint direction $p = G_q^* u / \|G_q^* u\|$ maximizes pairing with u on the unit write ball, when the adjoint image is nonzero. Its output column is $G_q G_q^* u / \|G_q^* u\|$, rather than u in general. A residual-aligned repair therefore requires solving the bounded realization problem and checking its residual. $\square$

One may equivalently solve for the unscaled column and multiply all realization errors by ρ .

Theorem 7.5 (A stable approximate repair). *Let $\Delta = cd$, $c > 0$, $\|d\| = 1$. At the post-latch start θ_0 , suppose there is a parameter direction e with $\|e\| \leq 1$ and*

$$\|\rho d - J(\theta_0)e\| \leq \varepsilon.$$

On a certified region assume $\|J(\theta) - J(\theta_0)\| \leq \omega$. If

$$2(\tau + \varepsilon + \omega) < \rho, \quad (11)$$

the loaded invariant holds at every point of the region. Each point admits a lift with residual at most $c/2$ and speed at most c/τ .

Proof. For every point in the certified region,

$$\begin{aligned}\|\rho d - J(\theta)e\| &\leq \|\rho d - J(\theta_0)e\| + \|(J(\theta) - J(\theta_0))e\| \\ &\leq \varepsilon + \omega.\end{aligned}$$

For a unit covector u , split its pairing with ρd into the actual column and its error. Adjointness gives

$$\rho |\langle d, u \rangle| \leq |\langle J(\theta)e, u \rangle| + \varepsilon + \omega \leq \|J(\theta)^*u\| + \varepsilon + \omega.$$

Thus any weak unit u satisfies $\rho |\langle d, u \rangle| \leq \tau + \varepsilon + \omega < \rho/2$. Since $\Delta = cd$ and $\|\Delta\| = c$, $4|\langle \Delta, u \rangle|^2 < \|\Delta\|^2$. Theorem 7.2 gives the pointwise lift and velocity bounds. Protection is established only over the certified region of this pass. No claim that the column remains fixed after later training is needed. $\square$

An available Lipschitz bound K on a ball of radius a supplies $\omega \leq Ka$. This smallness condition is part of the promise, and can fail even when the initial programming residual is small.

Theorem 7.6 (Finite numerical pass). *Let F have derivative J , K -Lipschitz on the closed ball $B = \overline{B}(\theta_0, a)$. Put $c = \|y^* - F(\theta_0)\|$. Suppose the actual step velocities $\widehat{v}_i$ satisfy on B*

$$\|\widehat{v}_i\| \leq W \leq a, \quad \|J\widehat{v}_i - \Delta\| \leq c/2 + \eta.$$

If $N \geq 1$ and

$$KW^2/N + \eta \leq c/4, \tag{12}$$

the N updates $\theta_{i+1} = \theta_i + \widehat{v}_i/N$ remain in B and their endpoint has residual at most $3c/4$. For an exact lift v and $\|\widehat{v} - v\| \leq \zeta$, $\|J\| \leq M$, sufficient velocity contracts are

$$W = c/\tau + \zeta, \quad \eta = M\zeta.$$

Rounding of a whole update is included by adding N times its parameter error to the velocity error.

Proof. Let $h = 1/N$. Assuming the first i steps are in the ball, their speed bound gives

$$\|\theta_i - \theta_0\| \leq ihW \leq W \leq a.$$

Induction proves domain safety for every iterate. Each update segment is in the ball by convexity.

The derivative-Lipschitz remainder estimate is

$$\|F(x + hv) - F(x) - hJ(x)v\| \leq K(hW)^2.$$

This uses a conservative constant; the usual $K/2$ estimate would improve the budget but is unnecessary. Adding the lift defect gives

$$\|F(\theta_{i+1}) - F(\theta_i) - h\Delta\| \leq K(hW)^2 + h(c/2 + \eta).$$

Sum the N inequalities and telescope. Since $Nh = 1$ and $F(\theta_0) + \Delta = y^*$,

$$\|F(\theta_N) - y^*\| \leq KW^2/N + c/2 + \eta \leq 3c/4.$$

If v is an exact lift and $\|\widehat{v} - v\| \leq \zeta$, then

$$\|\widehat{v}\| \leq c/\tau + \zeta, \quad \|J\widehat{v} - \Delta\| \leq \|Jv - \Delta\| + \|J\|\zeta \leq c/2 + M\zeta.$$

For an implemented update with error r relative to $x + v/N$, the effective velocity is $v + Nr$. This explains the rounding contribution stated in the theorem. $\square$

Corollary 7.7 (One realized repair supplies a finite pass). *Suppose Theorem 7.5 holds on the ball, J is K -Lipschitz there, $c/\tau \leq a$, and $K(c/\tau)^2/N \leq c/4$ for an integer $N \geq 1$. Then pointwise lifts can be selected and integrated by N finite steps to reach residual at most $3c/4$ while remaining in the ball. No Lipschitz assumption on the selected lifting field is required for this discrete result.*

Proof. At each point of the ball Theorem 7.5 supplies a lift with residual at most $c/2$ and speed at most c/τ . Select one at each point, defining it arbitrarily outside the ball. Apply Theorem 7.6 with $W = c/\tau$ and $\eta = 0$. Its domain induction and remainder estimate require only the stated pointwise bounds, rather than regularity of the selected lift. This proves a real-valued finite pass construction. For rational execution, a certified approximate backend must additionally satisfy the inexact contracts; classical pointwise selection alone does not give a bit-efficient implementation. $\square$

7.1 A concrete pass schedule

The region and numerical budgets can be stated as scalar acceptance tests. For a nonzero pass residual c , choose a ball radius $a = 2c/\tau$ and an initial repair error ε . A sufficient stability test is

$$2(\tau + \varepsilon + 2Kc/\tau) < \rho.$$

If the backend's total velocity error satisfies $\zeta \leq c/\tau$ and $M\zeta \leq c/8$, choose

$$N = \max\{1, \lceil 32Kc/\tau^2 \rceil\}.$$

Then $W = 2c/\tau$ and the numerical budget of Theorem 7.6 holds. This schedule needs polynomially many inner steps when Kc/τ^2 is polynomially bounded. It is conservative and provides an explicit certified implementation target; the archived adaptive Gauss–Newton loop uses a different schedule.

Verification of the schedule. The ball gives $\omega \leq Ka = 2Kc/\tau$. The velocity error bound gives $\|\hat{v}\| \leq c/\tau + \zeta \leq 2c/\tau = a$, so discrete domain safety applies. Finally,

$$\frac{K(2c/\tau)^2}{N} = \frac{32Kc/\tau^2}{N} \frac{c}{8} \leq \frac{c}{8}.$$

Adding $M\zeta \leq c/8$ proves the quarter-residual numerical budget. The norm bound M and the preactivation and state enclosures must be valid on the chosen ball, and programming must be included in the post-latch region certificate. $\square$

Assumption 7.8 (Certified pass and backend class). *An accepted encoded instance supplies at most $3(b + \ell)$ passes with:*

- (i) *sound DEQ well-posedness, evaluation and derivative certificates on the entire pass region, and $C^{1,1}$ bounds there;*
- (ii) *either the loaded invariant on that region or one unused programmable channel whose post-latch column satisfies Theorem 7.5;*
- (iii) *actual finite updates satisfying Theorem 7.6, with their decoded endpoint identified with the backend output;*
- (iv) *an encoding of size at most S per backend query and precision at most Q , where S, Q are polynomial in instance length and b ;*

(v) a selected complete pass backend, including programming, certificate checking, inference, linear algebra and rounding, with bit cost at most $A(S + Q + 1)^d$ for uniform family constants A, d .

Region bounds may be supplied as validated analytic bounds or enclosures; testing a finite list of successful solves does not establish them. The assumptions specify a reduction to an efficient certified backend, not a universal verifier for arbitrary smooth maps.

Theorem 7.9 (Certified training with budgeted termination). *For an instance in Assumption 7.8 with initial residual at most 2^ℓ , at most $3(b + \ell)$ completed passes produce the encoded output (2). The number of consumed channels is at most $3(b + \ell)$, and it suffices to provision that many channels. The total bit cost of the accepted encoded run is at most*

$$3(b + \ell)A(S + Q + 1)^d.$$

Thus the trainer is polynomial in instance length and requested bits when the displayed budgets, including ℓ , are polynomially bounded. If pass k has residual c_k , travel at most c_k/τ , and each latch costs at most ρ parameter travel, then the total travel is at most

$$4c_0/\tau + R_{\text{used}}\rho.$$

Programmed-vector changes and rounding add their separately certified travel budgets. A global domain ball is sufficient only if it contains this full travel budget and the pass regions used for certification.

More generally, execute the trainer through Theorem 4.8, with successful-answer predicate $\|F(\text{decode}(y)) - y^*\| \leq 2^{-b}$, sound acceptance, and the stated all-input transition-cost bound. Every run then terminates with certified success or inconclusive within the polynomial wrapper budget. For a fixed backend family satisfying Assumption 7.8, choose the budget to cover its complete encoded trace and dispatch; the wrapper preserves the above success, pass, and channel guarantees. Failure of a certificate or exhaustion of the budget is inconclusive, not a certificate of superpolynomial training complexity.

Proof. Pass activation preserves the output, so the residual used in the pass is also the residual of the preceding encoded endpoint. By the finite pass theorem, $c_{k+1} \leq (3/4)c_k$ whenever a nontrivial accepted pass is executed. An already successful endpoint can be retained with no further latches. Inductively,

$$c_k \leq (3/4)^k c_0.$$

Because $(3/4)^3 = 27/64 \leq 1/2$,

$$c_{3(b+\ell)} \leq 2^{-(b+\ell)} 2^\ell = 2^{-b}.$$

This argument requires contracts for only the finite run, not an infinite extension of it. Each pass consumes at most one distinct unused channel, so summing the latch count gives at most $3(b + \ell)$.

The encoded backend run is defined recursively by applying the selected pass backend to its preceding encoded output at the prescribed precision. The pass certificate identifies that output's decoded parameter with the finite numerical endpoint, tying convergence to the same calls whose work is counted. For every call, its complete bit cost is bounded by $A(S + Q + 1)^d$. Summing over $3(b + \ell)$ calls gives $3(b + \ell)A(S + Q + 1)^d$. The degree and coefficient are uniform for the chosen algorithm family. If the budgets are polynomially bounded in instance length and b , their composition is polynomial. Real-arithmetic operation counts alone would not justify this conclusion.

For the travel statement,

$$\sum_{k < K} c_k \leq c_0 \sum_{k < K} (3/4)^k \leq 4c_0.$$

Adding unit-time pass travel c_k/τ and the latch movements gives the displayed bound. If each approximate velocity additionally has a ζ_k error, unit-time pass travel additionally costs at most $\sum_k \zeta_k$. Programming vectors in the parameter metric similarly needs its own bound, even though their dormant output contribution is zero. Pointwise domain safety of an accepted pass does not automatically prove coverage of its entire certification ball by a previously chosen global domain ball. Finally apply Theorem 4.8 with the training answer predicate. Pass certificates justify acceptance on the original successful trace; the wrapper preserves it when the clock is sufficient. The all-input cost requirement also covers unsuccessful searches and checks, which Assumption 7.8 alone does not bound. $\square$

Remark 7.10 (Reserve width and computational size). *The guarantee counts channels, not stored scalar entries. Its bound is $O(b + \ell)$, independent of output dimension as a count, while gate realization and matrix-free products still depend on m and n . Programmable rank-one channels store $O(nR)$ vector entries. The theorem does not guarantee that a rank-one gate can realize arbitrary batch directions. For a fixed q , its image dimension is at most the write dimension; this restriction is not removed by increasing latch amplitude.*

7.2 Practical implementation: matrix-free loaded training

The regularized solve in the trainer can be implemented without assembling the batch training Jacobian or forming a matrix inverse. Its primitive operations are a Jacobian-vector product $v \mapsto Jv$ and an adjoint product $q \mapsto J^*q$. The operator passed to an iterative solver is

$$\mathcal{N}q = J(J^*q) + \tau^2 q, \quad \mathcal{N}\hat{w} \simeq \Delta, \quad \hat{v} \simeq J^*\hat{w}.$$

One application of $\mathcal{N}$ uses one product of each kind and vector operations. This replaces storage of an $m \times p$ Jacobian and an $m \times m$ normal matrix by product routines and solver workspace. The state solves, recurrence factors, batch states, and programmed channels still have their own storage costs.

Products through an equilibrium. For batch sample i , cache its accepted equilibrium x_i and activation derivative D_i , and set $E_i = I - D_i W_{\text{eff}}$. A parameter direction with induced perturbations $\delta W_{\text{eff}}, \delta U, \delta c, \delta w, \delta c_o$ gives

$$E_i \delta x_i = D_i(\delta W_{\text{eff}} x_i + \delta U u_i + \delta c), \quad (Jv)_i = x_i^T \delta w + \delta c_o + w^T \delta x_i.$$

For an output covector q , the corresponding adjoint state solve is $E_i^T \lambda_i = q_i w$. Contracting $D_i \lambda_i$ with the parameter directions gives the recurrent and input contributions to J^*q ; the readout contributions are $\sum_i q_i x_i$ and $\sum_i q_i$. These contractions use the structured weight parameterization, including the active bilinear channels, rather than a dense recurrent gradient. The factorized Jacobian reduction of Theorem 4.1 supplies state and adjoint solves through factorizations of the local blocks and the reduced matrix; no explicit inverse need be stored. Active channels must be included in those factors and conditioning certificates. The state Jacobian can be nonsymmetric, so conjugate gradients is used for the regularized normal system, not automatically for the state solve.

Iterative solution and conditioning. For $\tau > 0$, $\mathcal{N}$ is symmetric positive definite even when J is rank deficient. If $\|J\| \leq M$, its spectral condition number satisfies

$$\kappa(\mathcal{N}) \leq 1 + M^2/\tau^2.$$

Conjugate gradients (CG) therefore provides a standard matrix-free solver. In exact arithmetic, its classical estimate gives a sufficient iteration count of order $O(\sqrt{\kappa} \log(2 + \kappa \|\Delta\|/\xi))$ for normal residual tolerance ξ , with $\kappa = \kappa(\mathcal{N})$ [5]. A symmetric positive definite preconditioner may reduce this count; constructing and applying it are part of the backend cost. Regularization prevents singularity, but small τ can still cause expensive solves. Polynomial cost requires controlled conditioning, product precision, and encoded arithmetic, as in Assumption 7.8.

Acceptance with inexact products. The stopping test must bound the residual of the true normal equation. Suppose a checked product routine returns $\hat{a}$ with $\|\hat{a} - \mathcal{N}\hat{w}\| \leq \varepsilon_{\text{act}}$, and $\|\hat{\Delta} - \Delta\| \leq \varepsilon_{\Delta}$. Including arithmetic error in these enclosures gives the acceptance bound

$$\|\mathcal{N}\hat{w} - \Delta\| \leq \|\hat{a} - \hat{\Delta}\| + \varepsilon_{\text{act}} + \varepsilon_{\Delta} \leq \xi.$$

Proposition 7.3 then converts ξ into solution and velocity error. If the final adjoint product has error at most ε_v , the computed velocity satisfies

$$\|\hat{v} - J^*w\| \leq M\xi/\tau^2 + \varepsilon_v.$$

This quantity and whole-update rounding must fit the finite-pass budget of Theorem 7.6. A solver’s recursively updated residual alone does not account for inaccurate equilibrium states, derivative products, or rounding. Those errors need validated bounds, and approximate witnesses need the slack checks described after Proposition 7.3.

A practical execution policy. At each inner update, compute certified states, apply the two product routines in the regularized solver, check its true-residual enclosure, and form the accepted velocity. Warm starts and reuse of factorizations at the same parameter point can reduce work. A parameter change requires updated products or a certified bound for reusing them. Trial points must satisfy the inference and pass-region tests, and an unused channel is programmed only after its residual-aligned realization and stability checks succeed. Line searches and adaptive damping are useful practical choices, but their accepted updates must satisfy the finite-pass contract to retain the theorem’s guarantee. The loaded trigger requires no Lanczos margin monitoring or spectral decomposition.

The experiments in Section 10 already use matrix-free products and iterative normal solves; their product counts illustrate this implementation route. This discussion describes a practical candidate backend within the stated certification contracts.

8 An exponential obstruction to the global finite-pass certificate

The polynomial backend hypothesis is substantive even when inference is well conditioned and a repair is successfully realized. We construct a three-state tanh DEQ for which a pass certified by the single global curvature and speed budget of Theorem 7.6 requires exponentially many updates in the binary input length. The obstruction concerns that certificate and a specified trainable coordinate, not all algorithms for the training problem. In fact, a directly certified update solves the same interpolation task efficiently.

8.1 The training instance and successful repair

For an integer $L \geq 5$, put $\kappa = 2^{-L}$ and consider

$$x = \tanh \left(a\theta p q^T x + \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right), \quad p = \begin{pmatrix} 0 \\ 1 \\ \kappa^{-1} \end{pmatrix}, \quad q = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad F = x_2 + \frac{\kappa}{32} x_3.$$

There is one training sample, with target $c = 1/128$. All supplied coefficients are rational and their total binary encoding length is $\Theta(L)$. The large entry $\kappa^{-1} = 2^L$ requires only $L + 1$ binary digits. Initially $a = \theta = 0$. Program p, q while the channel is dormant, then activate $a = \kappa$ without changing the output. The backend subsequently fixes a, p, q , the readout and all other coefficients, and trains only θ . All parameter balls and derivatives in this section refer to this one-dimensional trainable slice.

Write $t = \tanh 1$, so $3/4 < t < 4/5$. These inequalities follow from $7 < e^2 < 9$. After activation, the unique equilibrium and output are

$$x_1 = t, \quad x_2 = \tanh(\kappa t \theta), \quad x_3 = \tanh(t \theta), \quad F_\kappa(\theta) = \tanh(\kappa t \theta) + \frac{\kappa}{32} \tanh(t \theta).$$

The initial residual is c . For example, the fixed accuracy request $b = 8$ asks for residual at most $c/2$. Thus the accuracy request need not grow with L in the lower bound below.

Proposition 8.1 (Conditioned inference and stable repair). *Inference has a unique equilibrium for every θ . At that equilibrium, $\|H_x^{-1}\| < 3$. On $|\theta| \leq R = 1/(8\kappa)$, the stable-repair theorem holds with*

$$\rho = \kappa, \quad \tau = \kappa/8, \quad \varepsilon = \kappa/4, \quad \omega = 3\kappa/64.$$

A derivative-Lipschitz bound on this ball is $K = \kappa/8$.

Proof. The equilibrium equations are triangular. The state Jacobian is $I - A$, where only $A_{21} = \kappa\theta \operatorname{sech}^2(\kappa t \theta)$ and $A_{31} = \theta \operatorname{sech}^2(t \theta)$ can be nonzero. Since $A^2 = 0$, its inverse is $I + A$. The inequality $|z| \operatorname{sech}^2 z \leq 1$ gives $\|H_x^{-1}\| \leq 1 + \sqrt{2}/t < 3$.

Let $J = F'_\kappa$. Before activation, the θ -derivative is zero; after activation $J(0) = (33/32)t\kappa$. In particular, $|\kappa - J(0)| \leq \kappa/4$. Furthermore,

$$0 \leq J(0) - J(\theta) = \kappa t \tanh^2(\kappa t \theta) + \frac{\kappa t}{32} \tanh^2(t \theta) \leq \kappa \left((\kappa \theta)^2 + \frac{1}{32} \right) \leq \frac{3\kappa}{64}.$$

Here $|\tanh z| \leq \min\{|z|, 1\}$ was used. With the unit parameter direction $e = 1$ and output direction $d = 1$, the stability test is $2(\tau + \varepsilon + \omega) = 27\kappa/32 < \rho$. Finally, differentiating again yields $|F''_\kappa| \leq 2\kappa^2 + \kappa/16 \leq \kappa/8$, since $\kappa \leq 1/32$. $\square$

8.2 An exponential lower bound for the certificate

Theorem 8.2 (Exponential globally certified pass length). *Let $L \geq 10$. Consider a pass from $\theta_0 = 0$ with fixed displacement $\Delta = c$, using N updates of size $\hat{v}_i/N$. Suppose a single ball $[-R, R]$, a valid Lipschitz constant K for J on that ball, a speed bound $W \leq R$, and an error budget $\eta \geq 0$ certify the pass through*

$$|\hat{v}_i| \leq W, \quad |J(\theta_i)\hat{v}_i - c| \leq c/2 + \eta, \quad KW^2/N + \eta \leq c/4.$$

Then $N \geq 2^{L-14}$. Consequently an implementation that executes these N updates requires $\Omega(2^L)$ operations. The bound applies regardless of how its lifted velocities or regularization thresholds are chosen.

Proof. The numerical budget gives $\eta \leq c/4$. At the first update,

$$W \geq \frac{c/2 - \eta}{J(0)} \geq \frac{c}{4\kappa},$$

because $0 < J(0) < \kappa$. Hence $R \geq 2$ for $L \geq 10$, and the interior of the certification ball contains $\theta_* = 1/t < 4/3$. Both terms of $F''_\kappa(\theta_*)$ are negative, so

$$K \geq |F''_\kappa(\theta_*)| \geq \frac{\kappa}{16} t^3 (1 - t^2) > \frac{243\kappa}{25600} > \frac{\kappa}{128}.$$

In particular $K, W > 0$, and the numerical budget implies $\eta < c/4$. Using $(c/2 - \eta)^2 - c(c/4 - \eta) = \eta^2 \geq 0$, we obtain

$$N \geq \frac{KW^2}{c/4 - \eta} \geq \frac{K(c/2 - \eta)^2}{J(0)^2(c/4 - \eta)} \geq \frac{Kc}{J(0)^2} \geq \frac{1}{16384\kappa} = 2^{L-14}.$$

Each explicitly executed update costs at least one operation. Since the instance length is $\Theta(L)$, this is exponential in that length. This conclusion does not cover algorithms that bypass the updates or use a different certificate. $\square$

The bound is not a consequence of deliberately overestimating curvature: the proof bounds every valid K from below using an actual second derivative inside the required ball. It also allows inexact velocities. A concrete admissible schedule is

$$K = \kappa/8, \quad W = R = 1/(8\kappa), \quad N = 2/\kappa = 2^{L+1}, \quad \eta \leq c/8.$$

Indeed, $KW^2/N = c/8$. The exact scalar Tikhonov velocity is $v = cJ/(J^2 + \tau^2)$ with $\tau = \kappa/8$; Proposition 8.1 and Theorem 7.2 supply its lift bounds. Since $|J| \leq \kappa$, velocity errors at most 2^{-12} fit both the displayed speed and defect budgets. Whole-update errors of order 2^{-L-20} contribute only a constant small velocity error after multiplication by N . Stored states have magnitude $O(2^L)$, so a dyadic implementation can use $O(L)$ bits at this fixed target accuracy. Activation evaluations at large arguments can use the bound $1 - |\tanh z| \leq 2e^{-2|z|}$ to certify saturation; other arguments require only precision growing polynomially with L . These observations describe a finite-precision realization, not a verified low-level implementation.

The sharper variation bound ω in Proposition 8.1 is essential. Substitution of the coarse bound KR into the extra sufficient stability test of Section 7.1 would reject this schedule: $KR = 1/64$ does not shrink with κ . The example therefore concerns the general finite-pass certificate, not an accepted execution of that more restrictive sufficient test.

8.3 A fast update and the scope of the obstruction

Proposition 8.3 (Directly certified interpolation). *For $L \geq 5$, the rational update $\theta = c/\kappa$ satisfies $|F_\kappa(\theta) - c| < c/3$, and hence meets the fixed accuracy request $b = 8$. Its parameter encoding has $O(L)$ bits.*

Proof. At this point $F_\kappa(\theta) = \tanh(tc) + (\kappa/32) \tanh(tc/\kappa)$. For $z \geq 0$, $\tanh z \geq z - z^3/3$, as follows by integrating $1 - \text{sech}^2 z = \tanh^2 z \leq z^2$. Consequently

$$F_\kappa(\theta) > 3c/4 - c^3/3 > 2c/3, \quad F_\kappa(\theta) < 4c/5 + \kappa/32 \leq 4c/5 + c/8 < c.$$

The stated error bound follows, and $\theta = 2^{L-7}$ is rational. $\square$

Thus this family is not an intrinsic hardness result for DEQ training, nor a reduction from the threshold-network problems of Blum and Rivest [11]. A global Taylor remainder budget combines a large parameter displacement with curvature concentrated near the start and can be exponentially pessimistic. Directional remainder certificates or local subdivision may avoid that expense. The exponentially large τ^{-1} and programmed-vector norm have short encodings; the complete pass backend just exhibited nevertheless violates the polynomial work promise in Assumption 7.8. The conditional training theorem is therefore unaffected.

9 Deep DEQ and feedforward extensions

9.1 A certificate class closed under composition

Define the inference class through recursively annotated architectures. Its leaves are certified implicit layers or explicit modules. Serial, parallel and residual nodes propagate domain, sensitivity and evaluation certificates; the implicit layers retain their architecture-specific reduced-margin certificates. Closure means closure of these annotated presentations with their quantitative budgets.

Proposition 9.1 (Serial implicit inference closure). *For a two-layer equilibrium stack $H_1(u, x_1) = 0$, $H_2(x_1, x_2) = 0$, its state Jacobian is*

$$\mathcal{J} = \begin{pmatrix} A & 0 \\ C & D \end{pmatrix}, \quad \mathcal{J}^{-1} = \begin{pmatrix} A^{-1} & 0 \\ -D^{-1}CA^{-1} & D^{-1} \end{pmatrix}.$$

If $\|A^{-1}\| \leq K_1$, $\|D^{-1}\| \leq K_2$ and $\|C\| \leq P$, then in the product max norm

$$\|\mathcal{J}^{-1}\| \leq \max\{K_1, K_2(1 + PK_1)\}.$$

Layerwise selected equilibria define a selected equilibrium of the stack; layerwise uniqueness gives stack uniqueness on the corresponding domain.

Proof. Solve $Av = r_1$ first, then $Dw = r_2 - Cv$. This gives the inverse formula and bounds $\|v\| \leq K_1 \|r\|$, $\|w\| \leq K_2(1 + PK_1) \|r\|$ in the product max norm. Sequential layer selection proves existence of the selected stack state. For uniqueness, the first layer fixes x_1 and the second then fixes x_2 . Domain inclusion must hold at each step. $\square$

For an inference chain of Lipschitz constants L_i , enclosed local output errors ϵ_i propagate as $e_i \leq L_i e_{i-1} + \epsilon_i$. Thus requested local precision must account for downstream amplification. Parallel nodes use block diagonal solves and sum their work; residual nodes add the explicit identity map. A polynomial-size tree has polynomial total inference cost when its propagated inverse, derivative, domain and precision budgets are polynomial and its leaf algorithms have polynomial costs with controlled output sizes. This is a closed certified subclass. A single feedback layer with fixed interface dimension is not closed under arbitrary stacking: the total interface dimension can grow, and a chain of bounded sensitivities can have exponentially large gain.

The input-sweep continuation and finite tracker are treated separately in Section 4.4. Saddle-coverage and reservoir-refresh certificates are not needed for this inference closure statement.

9.2 Training regularity of composed models

A finite stack of equilibrium modules supplies an output map after each module has its own sound inference certificate. The loaded training results then apply to the composed map. Polynomial depth alone does not bound propagated derivative constants; these must be certified.

Proposition 9.2 (Composition with trainable modules). *For $h(z, \vartheta) = g(f(z), \vartheta)$,*

$$Dh[v, w] = Dg[Df v, w].$$

Using the product max norm, joint derivative bounds L_f, L_g yield $\|Dh\| \leq L_g \max\{L_f, 1\}$. If the derivative Lipschitz constants are K_f, K_g and f is L_f -Lipschitz, a sufficient derivative variation bound is

$$K_h \leq K_g \max\{L_f, 1\}^2 + L_g K_f.$$

The direct downstream parameter contribution is included.

Proof. Let $\mathcal{L}(z, \vartheta) = (f(z), \vartheta)$. Its derivative is $D\mathcal{L}[v, w] = (Df v, w)$, whose operator norm in product max norms is at most $\max\{L_f, 1\}$. The chain rule gives $Dh = Dg D\mathcal{L}$ and the first-order estimate.

For two points, write the derivative difference as

$$Dg(\mathcal{L}z)D\mathcal{L}(z) - Dg(\mathcal{L}z')D\mathcal{L}(z') = [Dg(\mathcal{L}z) - Dg(\mathcal{L}z')]D\mathcal{L}(z) + Dg(\mathcal{L}z')[D\mathcal{L}(z) - D\mathcal{L}(z')].$$

The lift map is $\max\{L_f, 1\}$ -Lipschitz; its derivative variation is bounded by K_f . Taking norms gives $K_g \max\{L_f, 1\}^2 + L_g K_f$. In particular, an independent downstream parameter is passed through with derivative one. A product rule $L_g L_f$ alone can omit this contribution. $\square$

Equivalent Euclidean product norms introduce explicit norm-equivalence factors; they must be budgeted before applying the Hilbert-space training theorems. These compositional bounds concern smoothness, not adapter realizability or equilibrium uniqueness of the entire stack.

For feedforward networks, no equilibrium existence certificate is needed. Forward and backward products replace implicit sensitivity solves. The same programming, column stability and finite-pass hypotheses remain; placing adapters in an arbitrary layer does not automatically satisfy them. This extension uses the same abstract training lemmas and the parameter-aware composition rule rather than a separate training theory.

10 Numerical evidence and formal verification scope

A Python prototype (included in ancillary files) compares pseudo-arclength continuation with loaded Tikhonov loops on a compact digits 3 versus 8 task ($m = 10$, hidden width 7, eight base parameters, eight dormant channels, five seeds). The recorded aggregate values are:

method	success	median RMSE	mean latches	J -products	time (s)
pseudo-arclength	3/5	$3.3 \cdot 10^{-12}$	2.6	13273	8.6
loaded, fixed bank	5/5	$1.6 \cdot 10^{-13}$	4.6	820	0.9
loaded, programmable	5/5	$3.8 \cdot 10^{-13}$	6.4	700	1.1

Table 1: Recorded heuristic DEQ comparison on five digits-task seeds. Success means RMSE below 10^{-8} after final polishing.

The prototype maximizes witness pairing, and its equilibrium flag measures numerical convergence rather than uniqueness. The values illustrate the potential efficiency of a loaded trigger; they do not validate the realization and region contracts of Theorem 7.9.

Feedforward comparison. The prototyped feedforward comparison includes a no-reserve full-training control that succeeds on all five seeds. Both frozen-body adapter configurations succeed on one of five seeds and exhaust their reserves. This demonstrates why a reserve count must be distinguished from a quantitative gate-realization guarantee.

Selected-branch inference through two folds. The deterministic floating-point experiment follows the oriented equilibrium curve

$$H(u, x) = x - \tanh(2x + u) = 0$$

from $x = -4/5$ toward increasing x , through both simple folds, to the selected crossing of $u = -0.51$ beyond the second fold. It uses an adaptive pseudo-arclength predictor and a bordered Newton corrector. As a control, a monotone input sweep uses the same residual and warm-started Newton corrections but does not carry the branch orientation.

b	charts	corrections	selected residual	endpoint error	input-sweep branch error
8	35	31	$4.25 \cdot 10^{-6}$	$1.76 \cdot 10^{-5}$	1.77
16	35	53	$3.79 \cdot 10^{-10}$	$1.57 \cdot 10^{-9}$	1.77
24	35	69	$3.79 \cdot 10^{-10}$	$1.57 \cdot 10^{-9}$	1.77
32	35	81	$< 10^{-15}$	$< 10^{-15}$	1.77
40	35	91	$< 10^{-15}$	$< 10^{-15}$	1.77

Table 2: Floating-point continuation of the selected tanh branch. The chart count is independent of requested precision in this experiment, while corrector work grows with precision. The monotone input sweep ends at a different equilibrium despite an essentially zero terminal residual.

Every pseudo-arclength run crossed two sign changes of H_x . Along the $b = 32$ trace, the sampled full-row margin $\sqrt{H_x^2 + H_u^2}$ stayed above 0.448, the inverse norm of the unit-tangent bordered system stayed below 2.231, and $|H_x|$ fell to 0.0292 near a fold. The selected endpoint was $x = 0.787790669396$, whereas the monotone input sweep returned $x = -0.986134444248$. Thus residual accuracy alone did not establish the selected-branch answer predicate. The terminal reference value was computed independently by 80-digit decimal bisection on a fixed bracket. The tracker itself uses ordinary double precision, not directed interval arithmetic, so this is evidence about the mechanism, not a certified execution or evidence for the general polynomial-complexity theorem.

The quantitative statements above are machine-checked in Lean 4 at the level stated with each result, from the factorized kernel reduction and the training passes to the budgeted wrapper, the encoded tracker and scalar backend, the local chart layer, and the geometric and algorithmic content of the fold theorem. The correspondence between each statement and its checked declarations is recorded in source comments at the statement; the complete development is a single file in the ancillary material with a pinned toolchain, and its audited declarations use only the standard logical axioms. Results presented as paper proofs—the trial search, terminal monitor, and assembly theorem of Section 4.10, the arclength reparametrization and rational certificate synthesis of Section 4.8, backend cost summation, source-specific adapters, and the archived numerical runs—are not machine-checked.

10.1 Certified coverage and boundaries at a glance

Table 3 summarizes the accepted problem classes, the quantitative promises each route consumes, the guarantee it returns, and its cost regime. Every row is a conditional certificate: on instances outside

route	required promises	guarantee	bit cost
input sweep (Thms. 4.6, 4.7)	C^2 residual, certified start root, boundary exclusion, tube radius ρ with $\ H_x^{-1}\ \leq K$, $\ H_s\ \leq B$, second-derivative bound; polynomial encodings	unique selected branch tracked to 2^{-b}	polynomial in $L + b$
default continuation for certified solver classes (Thm. 4.11, Cor. 4.12)	instance certified for an established solver: contraction with gap γ in weighted, auxiliary, or logarithmic coordinates; certified evaluation and decoding	same equilibrium to 2^{-b} by the one continuation mechanism; explicit coverage of the quantified regimes of [17, 9, 18, 3, 30]	$O(\gamma^{-3}(p+1))$ solver calls; polynomial if γ^{-1} polynomial
outlined exceptions: direct solutions (Sec. 4.7)	separately certified class without a contraction gap, e.g. triangular ReLU with $A_{ii} < 1$	equilibrium by direct rational substitution, outside the continuation route	polynomial
fold continuation: new class (Thm. 4.14, Ex. 4.15)	row-rank margin σ , curve length L , C^2 bound, terminal bracket with $ du/d\ell \geq \beta$; chart-validation contracts	selected branch through simple folds to 2^{-b} , where no inherited contraction certificate exists on the route; $O(1+L/r)$ charts	polynomial
automatic charts (Lem. 4.18, Thm. 4.22)	certified seed and orientation; certified activation evaluators; neighborhood, margin, variation, and length bounds; terminal monitor with margins τ, β	self-generated chart path; selected equilibrium to 2^{-b} or inconclusive	polynomial given an inverse-polynomial progress scale
block elimination (Sec. 4.6)	permuted block-triangular tanh DEQ with per-block margins	equilibrium to 2^{-b} even when the full Euclidean inverse bound is $2^{\Theta(L)}$	polynomial precision overhead
loaded training (Thm. 7.9)	gate realization and column stability on pass regions; well-posed inference; finite-update budgets	interpolation to 2^{-b} in $O(b+\ell)$ passes	polynomial
global finite-pass certificate (Thm. 8.2)	single ball, curvature, and speed budget on the $\kappa = 2^{-L}$ three-state instance	certificate forces $N \geq 2^{L-14}$ updates despite conditioned inference and a successful repair	exponential for this certificate
chart-count boundary (Sec. 4.10)	polynomial encoding length without a positive progress scale	soundness retained; completeness may need exponentially many charts	possibly exponential

Table 3: Certified coverage and proved boundaries. Promises are inputs to the certificates; no row asserts unconditional convergence, and every route reports certified success or inconclusive within its budget.

its promise class the budgeted wrapper of Theorem 4.8 still terminates, returning inconclusive rather than an uncertified answer. The final two rows record proved boundaries, not failures of soundness: the finite-pass lower bound concerns one specific global certificate, and the same instance is solved directly by Proposition 8.3 in $O(L)$ bits.

11 Conclusion

We have developed a certified continuation framework for equilibrium computation and interpolation training in DEQs under quantitative certificate and backend promises. The operational outcome is certified success or inconclusive termination within a fixed polynomial budget. Supplied old inference certificates retain zero inference repairs, while sequential block elimination adds explicit coverage with only polynomial precision overhead despite large inter-block couplings. This extends the concrete uniform-inverse promise and uses established contraction ideas. The solver-inheritance theorem makes one continuation mechanism the polynomial default for explicitly quantified prior solver classes, while bordered continuation strictly extends coverage to a new class through simple folds under the additional length and terminal certificates. Theorems and lemmas in Section 4.10 further replace a supplied chart list by rational local generation and oriented assembly, conditional on

the seed, quantitative neighborhood bounds, and terminal-monitor contract. These new geometric results are paper proofs. Neither result asserts exhaustive coverage of every published convergence theorem. For inference, Theorem 4.1 (reduced denominator) exposes feedback singularities, and Theorem 4.2 (local transfer) provides conditioning bounds for the stated block structure. Theorem 4.6 (compact input sweep) continues a selected start root beyond the contractive setting. Its hypotheses include a bounded state region, exclusion of boundary zeros, and invertible state Jacobians at every zero. For numerical tracking, a certified tube must supply the uniform bounds

$$\|H_x^{-1}\| \leq K, \quad \|H_s\| \leq B, \quad \|D^2H\| \leq H_2.$$

Choose a tracking radius r inside that tube with $KH_2r \leq 1/4$ and $r \leq 2^{\ell_r}$, where ℓ_r is a nonnegative integer. The Theorem 4.7 (rounded tracker) permits the schedule

$$N = \max\{1, \lceil 2KB/r \rceil\}, \quad m = b + \ell_r + 2, \quad \delta \leq \min\{r/8, 2^{-(b+2)}\}.$$

The initial state must be within $r/2$ of the start root; each correction's solve and rounding errors must satisfy $K\eta + \zeta \leq \delta$. All corrections remain in their certified balls, and the computed endpoint has state error at most 2^{-b} . There are exactly Nm corrections. If each complete correction costs at most $A(S + P + 1)^d$, where S bounds query size and P bounds working precision, their total work is at most

$$N(b + \ell_r + 2)A(S + P + 1)^d.$$

Polynomial bounds on these quantities and on tube and stage certification give polynomial total inference cost; the tracker does not discover those certificates. Theorem 5.2 (concrete tanh DEQ) instantiates a bounded scalar DEQ with a rational activation evaluator and finite dyadic updates. Its work envelope is $O((b + 1)(S + b + 1)^{12})$, hence $O((S + b + 1)^{13})$, with S now denoting input fraction size. This concrete backend does not supply general multidimensional gate or training-region certificates.

For training, programmable dormant bilinear channels permit an output-preserving repair when a loaded Tikhonov pass fails. Theorem 7.2 (Tikhonov dichotomy), Theorem 7.5 (stable repair), and Theorem 7.6 (finite numerical pass) connect the diagnosis to a certified interpolation displacement. Theorem 7.9 (certified training) then reduces a residual bounded by 2^ℓ to 2^{-b} using at most $3(b + \ell)$ passes and channels, with total bit cost at most

$$3(b + \ell)A(S + Q + 1)^d.$$

Here S bounds pass query size, Q bounds precision, and A, d are fixed constants of the selected complete pass backend, including programming, certificate checking, inference, linear algebra, and rounding. The certified pass and backend assumptions require sound well-posedness and derivative bounds over the full pass regions, either a loaded invariant or a realizable stable repair, and actual finite updates meeting the numerical error budgets. When S, Q, ℓ and the complete backend budgets are polynomially bounded in instance length and b , the displayed cost is polynomial. The reserve bound counts channels; their programmed vectors have additional storage and realization costs. The inference and training constants A, d belong to their respective selected backend families and need not coincide.

The DEQ numerical comparison illustrates the potential computational benefit of a loaded trigger. The feedforward comparison also shows why a channel reserve alone does not ensure successful interpolation. These heuristic experiments do not validate the certified algorithm's full contracts. The formal verification scope describes the quantitative results checked in Lean and the implementation claims outside that scope. The matrix-free implementation discussion explains how

training can use implicit Jacobian and adjoint products, iterative regularized solves, and checked residuals without forming the training Jacobian or an explicit inverse.

The next practical step is to construct efficient gate and region certificates for larger DEQs. The present results identify exactly where those certificates enter the complexity guarantees, while the composition bounds extend the inference framework to finite stacks with separately controlled budgets.

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